



Siderophile and chalcophile elemental constraints on the origin of the Jinchuan Ni-Cu-(PGE) sulfide deposit, NW China

Xie-Yan Song^{a,*}, Reid R. Keays^b, Mei-Fu Zhou^c, Liang Qi^c,
Christian Ihlenfeld^{b,d}, Jia-Fei Xiao^a

^a State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences,
46th Guanshui Road, Guiyang 550002, PR China

^b School of Geosciences, Monash University, Australia and Department of Earth Sciences, Laurentian University, Sudbury, Canada

^c Department of Earth Sciences, The University of Hong Kong, Hong Kong, PR China

^d Anglo American Plc, 20 Carlton House Terrace, London SW1Y, 5AN, United Kingdom

Received 15 April 2008; accepted in revised form 20 October 2008; available online 6 November 2008

Abstract

The Jinchuan deposit, NW China, is one of the world's most important Ni-Cu-(PGE) sulfide deposits related to a magma conduit system and is hosted in an ultramafic intrusion. The intrusion is composed of lherzolite and dunite with the two largest sulfide ore bodies (named as ore body 1 and 2) in its middle portion. The sulfide ores may be disseminated, net-textured, or massive. The disseminated and net-textured sulfide ores are characterized by variable but generally low PGE concentrations: 10–3200 ppb Pt, 240–9800 ppb Pd, 17–800 ppb Ir, 25–1500 ppb Ru, and 15–400 ppb Rh in 100% sulfides. The massive sulfide ores are extremely low in Pt (<30 ppb) on a 100% sulfides and have very high Cu/Pd ratios, ranging from 10^4 to 4.5×10^5 . The low PGE contents suggest that the sulfide ores formed from the silicate magmas that had already experienced **prior-sulfide** separation.

Our calculations indicate that if the first stage basaltic magmas had contained 6.3 ppb Pt, 6.2 ppb Pd, and 0.1 ppb Ir, 0.008% sulfide removal would result in PGE-depletion in the residual magma with 0.57 ppb Pt, 0.25 ppb Pd, and 0.009 ppb Ir. The Jinchuan sulfides were formed by a second stage of sulfide segregation from a PGE-depleted magma under silicate/sulfide liquid ratios (R-factor) ranging from 10^3 to 10^4 in a deep-seated staging chamber. The massive sulfide ores and some of the net-textured sulfide ores exhibit strong negative Pt-anomalies that cannot be explained by sulfide segregation under variable R-factors. Instead, the sulfide melts that formed the massive ores were segregated from magmas experienced prior fractionation of Pt–Fe alloy. Alternatively, the Pt may have been selectively leached by hydrothermal fluids during remobilization of the sulfide melts that produced the massive sulfides, which occur in cross-cutting veins. We propose that the Jinchuan intrusion and ore bodies were formed by injections of **sulfide-free and sulfide-bearing** olivine mushes from a deep-seated staging chamber.

© 2008 Elsevier Ltd. All rights reserved.

1. INTRODUCTION

Magma conduit systems provide very important environments for the formation of magmatic Ni-Cu-(PGE) sulfides deposits, such as the giant Noril'sk-Talnakh deposit,

Russia and Voisey's Bay deposit, Canada (Naldrett et al., 1995; Lightfoot and Naldrett, 1999; Li and Naldrett, 1999; Lightfoot and Keays, 2005). These studies have shown that the concentrations of siderophile and chalcophile elements in the sulfides of these deposits are closely dependent on the compositions and evolution of the parental magmas and the processes of sulfide segregation in magma conduit systems. The Jinchuan Ni-Cu-(PGE) sulfide deposit in Gansu Province, northwestern China, contains

* Corresponding author. Fax: +86 0851 5895538.

E-mail addresses: songxieyan@vip.gyig.ac.cn (X.-Y. Song), cihlenfeld@angloamerican.co.uk (C. Ihlenfeld).

5.45 million tonnes of metallic Ni and is hosted in a small ultramafic body with less than 1.5 km² surface expression. About 85% metallic reserves are contained in three giant ore bodies in the middle portions of the Jinchuan intrusion (Sixth Geological Unit, 1984). The origin of such giant size ore bodies in such a small intrusion is still enigmatic. Tang and Li (1995), De Waal et al. (2004), and Song et al. (2006) proposed that the Jinchuan intrusion and sulfide ore bodies were formed by injection of pulses of sulfide-free and sulfide-bearing crystal-mushes from a magma chamber at a lower level.

The Jinchuan sulfide ores are particularly poor in PGE relative to those of the Noril'sk deposit on a 100% sulfide basis (Naldrett, 2005). Chai and Naldrett (1992a) also noted that there are large compositional variations between the different Jinchuan ore bodies. However, how such a large amount of sulfides was concentrated in the small Jinchuan ultramafic intrusion and what processes were responsible for the low PGE contents of the sulfide ores are still not well understood. This paper presents new lithological observations together with new major element oxide, Ni, Cu, and PGE data for whole-rock samples from a surface traverse as well as from two drill holes across the middle portion of the Jinchuan intrusion and the largest Ni-Cu-(PGE) sulfide ore bodies. This new dataset confirms that the Jinchuan sulfides are low in PGEs and in particular Pt. Our study suggests that the concentrations of siderophile and chalcophile elements of the Jinchuan sulfides

are not only dependent on the sulfide segregation in a deep-seated staging magma chamber in a magma conduit system, but also are constrained by sulfide removal from the silicate magmas before they intruded the staging magma chamber. In addition, either fractionation of Fe-Pt alloys from the silicate magmas prior to the sulfide segregation or loss of Pt from the sulfides during late hydrothermal remobilization may have produced the negative Pt-anomalies of the Jinchuan sulfide ores. This study provides new insights into the understanding of the processes in a magma conduit system and its relationship to the formation of Ni-Cu-(PGE) sulfide deposits.

2. GEOLOGICAL BACKGROUND

2.1. Regional geology

The North China Craton is bounded to the north by the Paleo-Central Asia Orogenic Belt and by the Central Orogenic Belt to the south (Fig. 1). The Central Orogenic Belt consists of the Qilian and Kunlun Paleozoic Orogenic Belts in the west and the Qinling-Dabie Mesozoic Orogenic Belt in the east (Yang et al., 2005). The southwestern margin of the North China Craton is composed of the Longshoushan Terrane, which is bounded by two regional thrust faults on its northern and southern sides, and the early Paleozoic Qilianshan orogenic belt, which occurs to its south (Sixth Geological Unit, 1975, 1984). In the Longshoushan Terrane, the

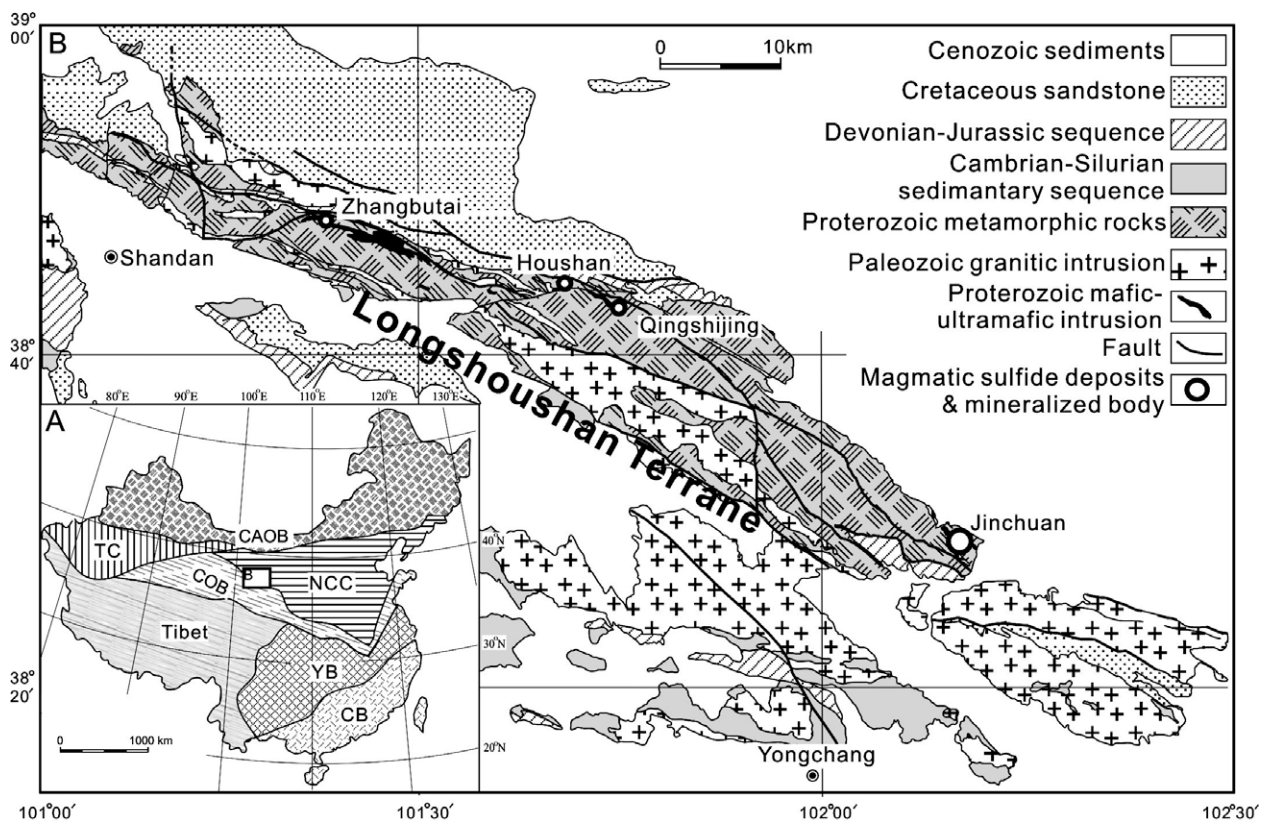


Fig. 1. (A) Tectonic units of China. CAOB, Central Asian Orogenic Belt; NCC, North China Craton; TC, Tarim Craton; COB, Central Orogenic Belt; YB, Yangtze Block; CB, Cathaysian Block. (B) Simplified regional geological map showing locations of the Jinchuan intrusion and several other mafic-ultramafic intrusions in the Longshoushan Terrane, which is located at the southwestern margin of the North China Craton.

Paleoproterozoic and Mesoproterozoic metamorphic units are unconformably overlain by Neoproterozoic and Paleozoic conglomerate, sandstone, and limestone. There are several mafic–ultramafic intrusions at the northern part of the terrane, including the Jinchuan and Yejili intrusions (Fig. 1) (Sixth Geological Unit, 1975, 1984; Tang, 1991; Barnes and Tang, 1999).

Tang and Li (1995) dated the Jinchuan intrusion by Sm–Nd isotope at 1508 Ma. Using Re–Os isotope age dating, Keays et al. (2004) reported that the Jinchuan intrusion had formed at 1332 Ma and then underwent a thermal event at 877 Ma. Zhou et al. (2002) concluded that it had been formed during a Proterozoic rifting event. Recent U–Pb SHRIMP dating of zircon and baddeleyite indicate that the Jinchuan intrusion formed at 827 ± 8 Ma and is related to a mantle plume, whose emplacement resulted in the break-up of the Rodinia super continent (Li et al., 2005).

2.2. Field relationship and petrography

The NW–SE striking elongate Jinchuan intrusion is 6.5 km long and a few tens to ~ 500 m wide. It was emplaced in the Mesoproterozoic metamorphic rocks and dips to the southwest with oblique angles varying from 50° to 80° (Fig. 2). The Jinchuan intrusion comprises mainly lherzolite and dunite and is divided into four segments, signed as I to IV, by strike slip faults called F_8 , F_{16} , and F_{23} , respectively (Fig. 2A) (Sixth Geological Unit, 1984). Segment II is well exposed in the middle portion of the Jinchuan intrusion between the faults F_{16} and F_{17} and extends downwards along dip to more than 1000 m in its middle part. Segment I is exposed in the west of Segment II, whereas the unexposed Segments III and IV occur at the western and eastern terminations of the Jinchuan intrusion, respectively.

The Jinchuan intrusion contains more than 100 sulfide ore bodies. The largest ore bodies named as No. 1, No. 2, and No. 24 (Fig. 2) contain 85% of the total Ni, Cu, and PGE reserves of the Jinchuan intrusion (Sixth Geological Unit, 1984). The No. 1 and No. 2 ore bodies are hosted in the western and eastern portions of Segment II, respectively, whereas the No. 24 ore body is situated at the base of Segment I (Fig. 2).

Dunite overlies lherzolite in the middle and eastern portions of Segment II. There are sharp boundaries between the dunite and lherzolite (Fig. 2). The dunite has high olivine contents (80–95 modal%) and relatively small olivine grain sizes (0.5–3 mm in diameter). In contrast, the lherzolite comprises 50–75 modal% olivine, 25–45% pyroxene and has large olivine grain sizes (2–6 mm). Lherzolites containing more than 10 modal% plagioclase are termed plagioclase lherzolites (Fig. 2A). Coarse-grained pyroxenite occurs along the margins of Segment II.

The western portion of Segment II has a core of net-textured sulfide-dunite with a thin margin of disseminated sulfide-dunite, which in turn is enveloped by the dominant coarse-grained lherzolite (Fig. 2). The net-textured and disseminated sulfides comprise the No. 1 ore body, which is 1500 m long along strike and extends from a depth of 200 m to >1100 m below the surface and accounts for more

than 50% of the total reserves of Ni, Cu, and PGE of the Jinchuan deposit (Fig. 2) (Sixth Geological Unit, 1984). To the east, disseminated and net-textured sulfide-dunites occur in the lower part of the segment and comprise the No. 2 ore body, which is ~ 1300 m long and overlain by coarse-grained lherzolite (Fig. 2). Bean pod-shaped or vein-like massive ore bodies occur along fractures within and at the margins of the No. 2 ore body, or in the country rocks to the Jinchuan Intrusion. The massive sulfides have sharp boundaries with their host rocks and contain various inclusions of ultramafic and metamorphic rocks (Sixth Geological Unit, 1984).

The disseminated and net-textured sulfide ores in the Jinchuan intrusion generally contain 5–20 modal% and 25–40 modal% sulfides, respectively, whereas the massive ores contain >70 modal% sulfides. The most common sulfides are pyrrhotite, pentlandite, and chalcopyrite (Sixth Geological Unit, 1984).

3. SAMPLING AND ANALYTICAL METHODS

Segment II and No. 1 and No. 2 ore bodies in the middle portion of the Jinchuan intrusion are sampled. Twelve samples (JC05-1 to JC05-12) of fine-medium-grained dunite, medium-coarse grained lherzolite and pyroxenite were collected along surface traverse A–A' (Fig. 2). Twenty-four samples were collected from the drill hole ZK83 (JD04-102 to JD04-127) and 40 samples were collected from the drill hole ZK86 (JD04-61 to JD04-101) that intersected No. 1 and No. 2 ore bodies in the western and eastern portions of Segment II, respectively (Fig. 2). The drill hole samples include dunite, lherzolite, sulfide-lherzolite, and sulfide-dunite. The major oxide and PGE compositions of these samples are listed in Table 1.

Samples were powdered using an agate mill and then digested with 1 ml of HF and 0.5 ml of HNO_3 in screw top PTFE-lined stainless steel bombs at 190°C for 12 h. Nickel, Cu and the major oxides except SiO_2 were measured by Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES) at the Guangzhou Institute of Geochemistry, Chinese Academy of Sciences. The precision of the analyses were generally better than 1% when the contents of the elements were greater than 200 ppm, and 1–3% when less than 200 ppm. International reference materials PCC-1 (dunite) and W-2 (basalt) were used for calibration. Whole-rock SiO_2 for the rocks and ores were determined using a gravimetric method. Whole-rock S contents for the ores and sulfide-poor rocks were obtained using gravimetric method and IR-absorption spectrometry, respectively. We used RTS-2 (Sulfide Tailings by Canadian Certified Reference Materials Project), which contains 18.95 ± 0.37 wt% S, as reference materials for ores, and used DZE-1 (ultramafic rocks by Xian Institute of Geological and Mineral Resources, Chinese Academy of Geological Science), which contains 510 ppm S, as reference materials for sulfide-poor rocks. The detection limit of the IR-absorption spectrometry is about $1 \mu\text{g/g}$ and the accuracies are estimated to be better than $\pm 8\%$. The PGE analyses were done by ID-ICP-MS digested with HF and aqua regia combined with sodium peroxide fusion and Te co-precipitation at the

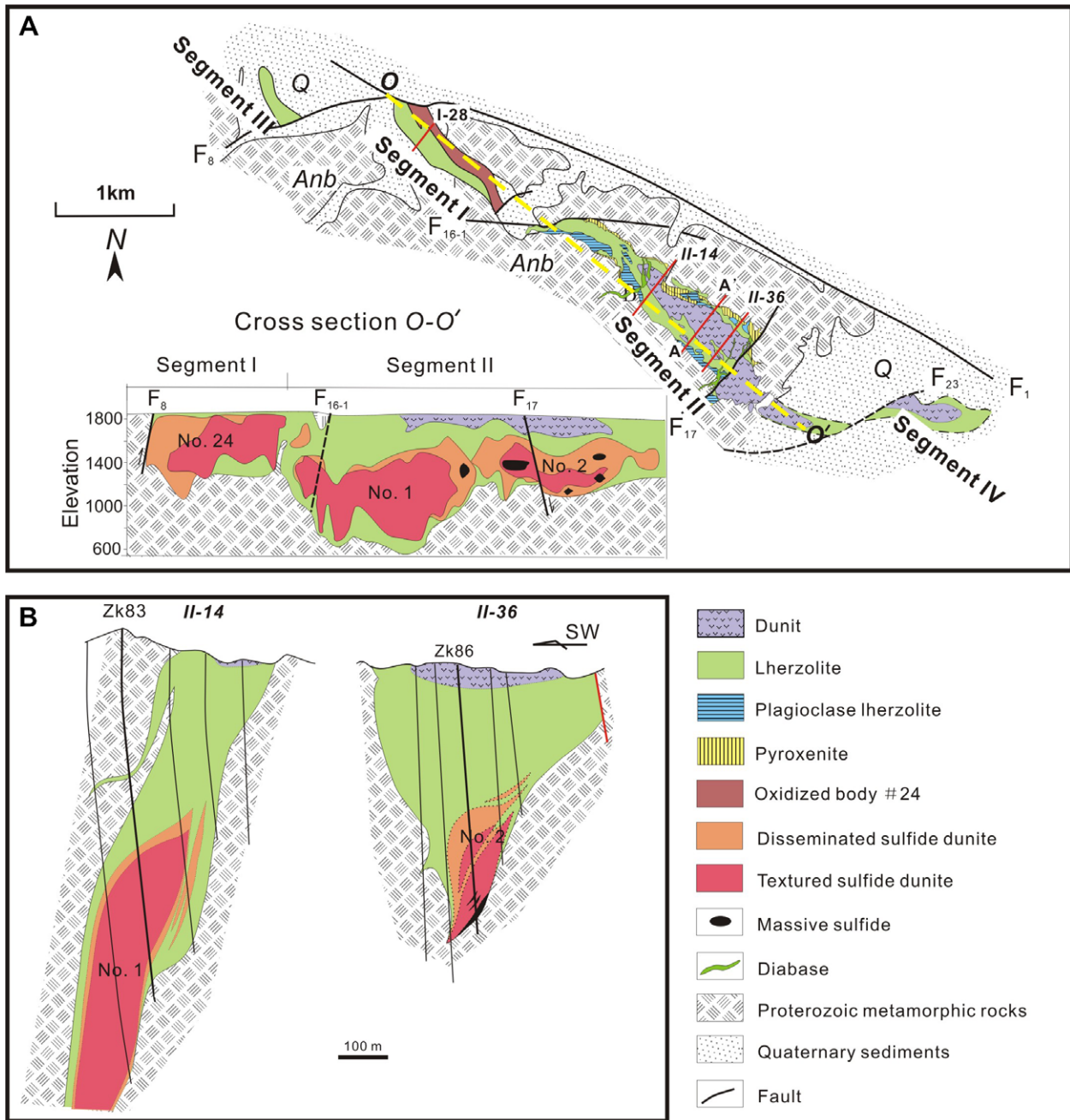


Fig. 2. Geological map of the Jinchuan intrusion and cross sections of Segment II. (A) Geological map and a cross section of the Jinchuan intrusion, showing distribution of rocks and the largest ore bodies. (B) Cross section of exploration lines II 14 and II 36 in Segment II showing sampled bore holes.

Chinese Academy of Geological Sciences. A detailed description of the method for the PGEs is presented by Qi et al. (2004) and Qi and Zhou (2008). Precision and accuracy as demonstrated by analyzing reference materials, such as UMT-1 and WPR-1, are better than 10% (Table 2). Four of the massive sulfide samples were analyzed at the Geoscience Laboratories in Sudbury, Canada, in order to confirm the very low Pt contents of the massive sulfide samples. Details of the analytical method used at the Geoscience Labs are presented in Keays and Lightfoot (2007). The accuracy of PGE data from the Geoscience Labs are also better than

10% and the agreement in the PGE analyses done at the two labs was excellent. The procedural blank for Pt is generally <0.xx ppb.

4. ANALYTICAL RESULTS

4.1. Major oxides

Most of the Jinchuan samples have large losses on ignition (LOI) because of alteration and high sulfide contents (Table 1). Thus, the weight percentages of the oxides of

Table 1
Concentrations of major oxides and trace elements in the Jinchuan rocks and ores.

Location	II-14-ZK83													
Rock type	Lh	Lh	Lh	Lh	NT-SD	DS-SD	NT-SD	NT-SD	NT-SD	NT-SD	NT-SD	NT-SD	NT-SD	NT-SD
Depth (m)	424	447	459	475	490	513	560	590	610	630	648	660	681	710
Sample	JD04-127	JD04-110	JD04-126	JD04-124	JD04-119	JD04-121	JD04-120	JD04-123	JD04-118	JD04-117	JD04-116	JD04-115	JD04-22	JD04-114
<i>Oxides in wt%</i>														
SiO ₂	42.4	44.3	43.1	43.9	29.5	41.9	29.1	32.4	30.3	30.3	29.8	30.8	33.8	34.9
TiO ₂	0.77	0.96	0.40	0.40	0.08	0.38	0.09	0.10	0.08	0.07	0.10	0.05	0.07	0.08
Al ₂ O ₃	10.0	6.29	6.59	9.02	1.00	5.09	1.20	2.16	2.12	1.37	0.57	0.86	1.54	1.59
Fe ₂ O ₃ (t)	7.1	13.9	12.7	11.0	23.4	13.5	24.4	19.4	18.9	21.2	23.4	23.2	20.2	20.5
MnO	0.06	0.22	0.18	0.15	0.14	0.17	0.12	0.14	0.15	0.14	0.15	0.14	0.14	0.15
MgO	17.3	26.3	26.7	23.9	26.8	29.5	28.3	27.7	31.7	30.5	28.6	28.8	29.5	30.1
CaO	14.4	5.38	5.60	5.95	0.56	3.58	1.37	1.80	1.53	0.71	0.19	0.38	0.83	0.84
Na ₂ O	1.22	0.83	0.80	1.38	0.05	0.55	0.03	0.15	0.13	0.06	0.03	0.06	0.17	0.16
K ₂ O	0.71	0.31	0.57	0.59	0.03	0.39	0.01	0.07	0.15	0.07	0.05	0.04	0.16	0.22
P ₂ O ₅	0.05	0.14	0.04	0.05		0.03								
S	0.07	0.29	0.12	0.14	8.92	0.31	9.0	5.85	4.60	6.20	7.76	7.65	5.38	5.19
LOI	2.61	2.21	2.70	4.75	11.0	5.38	9.20	9.50	8.60	9.30	10.1	9.80	9.20	8.80
Total	96.7	100.8	99.5	101.1	92.6	100.5	93.8	93.4	93.6	93.8	93.0	94.1	95.6	97.4
<i>Trace elements in ppm</i>														
Ni	84	915	907	881	24989	1740	22219	9509	13632	17222	18400	20213	15977	12629
Cu	44	59	34	65	15892	173	4133	13136	2113	4190	15128	2439	4170	1331
<i>Platinum-group elements in ppb</i>														
Ir	0.202	0.417	0.148	0.302	115.5	1.56	124.8	45.03	29.8	20.0	15.6	12.0	8.70	14.3
Ru	0.881	0.850	0.392	0.745	281.4	3.13	222.9	91.20	44.4	21.9	16.4	13.6	9.11	14.4
Rh	0.113	0.101	0.366	0.128	78.4	0.995	59.66	23.40	15.4	13.6	12.4	11.8	7.28	7.86
Pt	2.22	18.7	1.35	3.68	279	10.1	36.4	57.7	60.6	60.1	52.0	54.3	96.7	118
Pd	4.91	15.0	4.11	6.02	264	15.7	79.7	290	377	525	482	441	412	86.3
Pd/Ir	24.3	36.0	27.7	19.9	2.3	10.1	0.6	6.4	12.7	26.3	30.9	36.8	47.3	6.0
Cu/Pd	8880	3912	8275	10730	60294	11078	51861	45364	5607	7982	31403	5527	10125	15430
Location	II-14-ZK83										II-36-ZK86			
Rock type	NT-SD	NT-SD	NT-SD	NT-SD	NT-SD	DS-SD	DS-SD	Lh	Lh	DS-SD	Dunite	Lh	Lh	
Depth (m)	735	760	773	780	785	800	807	840	870	871	12	49	66	
Sample	JD04-113	JD04-105	JD04-112	JD04-106	JD04-111	JD04-108	JD04-104	JD04-107	JD04-103	JD04-102	JD04-96	JD04-98	JD04-97	
<i>Oxides in wt%</i>														
SiO ₂	29.1	26.6	30.0	25.5	30.3	40.6	31.0	44.5	44.9	35.6	37.5	37.9	37.6	
TiO ₂	0.08	0.09	0.05	0.12	0.06	0.30	0.33	0.63	0.26	0.11	0.17	0.21	0.19	
Al ₂ O ₃	1.07	0.79	0.61	0.72	1.70	4.87	2.54	7.76	10.88	3.09	2.23	2.74	3.02	
Fe ₂ O ₃ (t)	22.3	24.4	19.5	24.8	21.5	15.2	23.7	11.7	11.0	16.1	13.6	12.8	12.5	
MnO	0.14	0.15	0.13	0.16	0.15	0.15	0.15	0.14	0.11	0.15	0.17	0.16	0.16	
MgO	28.3	28.2	25.4	27.9	29.3	28.2	24.2	21.1	19.2	31.9	34.7	33.5	33.4	
CaO	0.55	0.26	1.91	0.43	1.01	3.41	2.42	6.12	6.01	1.14	1.31	1.49	1.76	
Na ₂ O	0.06	0.02	0.04	0.02	0.14	0.24	0.16	0.57	0.64	0.27	0.11	0.19	0.13	
K ₂ O	0.06	0.01	0.01	0.01	0.11	0.36	0.13	0.61	4.64	0.34	0.03	0.13	0.13	
P ₂ O ₅						0.02		0.08	0.02	0.01	0.02	0.03	0.02	
S	7.32	8.85	7.80	8.10	5.80	2.11	7.26	1.10	0.12	2.55	0.15	0.13	0.11	

LOI	10.10	11.10	12.17	11.20	9.01	5.71	7.74				11.78	9.33	10.09
Total	91.8	91.7	89.8	90.9	93.2	99.1	92.2	93.2	97.7	88.7	101.6	98.5	98.9
<i>Trace elements in ppm</i>													
Ni	25662	19594	18957	21331	18437	5281	18074	805	498	7493	1403	1202	1180
Cu	6354	24923	23370	15455	3782	1334	8605	194	119	1262	41.8	10.6	20.2
<i>Platinum-group elements in ppb</i>													
Ir	13.8	18.4	5.25	18.4	17.0	4.78	15.8	0.110	0.063	16.4	0.038	0.126	0.045
Ru	14.6	19.4	15.7	23.3	15.3	11.0	41.4	0.213	0.211	28.3	0.498	0.435	0.561
Rh	13.1	12.5	7.07	8.03	10.561	1.89	6.11	0.063	0.041	4.63	0.021	0.023	0.025
Pt	90.0	109	123	34.5	94.6	19.5	14.6	1.41	1.62	22.5	0.369	1.644	0.664
Pd	382	119	348	77.5	106	62.1	117	2.46	1.71	184	0.331	0.545	2.31
Pd/Ir	27.7	6.4	66.3	4.2	6.2	13.0	7.4	22.4	27.1	11.2	8.6	4.3	51.5
Cu/Pd	16613	209820	67103	199426	35845	21469	73643	78816	69430	6871	126122	19514	8739
Location	II-36-ZK86												
Rock type	Lh	Lh	Lh	Lh	Lh	Lh	Lh	Lh	Lh	Lh	DS-SD	DS-SD	DS-SD
Depth (m)	85	98	113	139	167	193	193	227	259	271	310	322	338
Sample	JD04-99	JD04-100	JD04-101	JD04-89	JD04-93	JD04-90	JD04-91	JD04-94	JD04-92	JD04-95	JD04-88	JD04-87	JD04-86
<i>Oxides in wt%</i>													
SiO ₂	37.0	36.5	37.7	38.2	37.7	38.0	37.0	37.9	39.3	39.2	38.6	38.5	36.2
TiO ₂	0.24	0.35	0.30	0.27	0.29	0.36	0.21	0.26	0.38	0.30	0.26	0.43	0.16
Al ₂ O ₃	2.95	3.20	3.28	3.57	3.31	3.13	2.79	3.28	2.17	3.00	2.94	2.82	3.07
Fe ₂ O ₃ (t)	13.0	13.5	13.9	13.4	13.1	13.4	12.6	12.9	13.3	13.5	16.8	16.9	19.3
MnO	0.17	0.14	0.18	0.19	0.17	0.18	0.17	0.16	0.18	0.17	0.17	0.16	0.15
MgO	33.3	32.5	35.0	35.6	33.1	34.1	34.0	34.2	35.0	34.5	32.0	31.2	31.6
CaO	1.41	1.40	0.98	2.22	1.84	2.21	1.53	1.93	1.14	1.61	2.34	1.75	2.33
Na ₂ O	0.15	0.13	0.08	0.31	0.10	0.32	0.12	0.20	0.11	0.21	0.29	0.33	0.41
K ₂ O	0.19	0.18	0.02	0.15	0.04	0.18	0.10	0.16	0.21	0.19	0.15	0.28	0.13
P ₂ O ₅	0.03	0.06	0.05	0.03	0.05	0.05	0.03	0.03	0.06	0.04	0.02	0.04	
S	0.16	0.34	0.37	0.34	0.15	0.19	0.44	0.28	0.19	0.80	2.41	2.88	4.06
LOI	10.9	10.7	9.94	8.06	10.9	10.7	9.05	9.67	6.83	8.24	6.74	7.8	6.8
Total	99.3	98.7	101.5	102.0	100.6	102.6	97.6	100.6	98.7	101.0			
<i>Trace elements in ppm</i>													
Ni	1201	1312	1312	1310	1260	1397	1238	1371	1690	1716	6019	5660	8291
Cu	29.3	16.6	15.7	18.6	26.3	131	30.6	33.4	178	322	1788	2808	8651
<i>Platinum-group elements in ppb</i>													
Ir	0.054	0.058	0.042	0.116	0.067	0.083	0.055	0.082	0.192	0.626	0.882	1.37	7.16
Ru	0.312	0.491	0.280	0.413	0.444	0.580	0.506	0.476	0.765	1.82	1.50	1.63	5.36
Rh	0.024	0.024	0.026	0.024	0.033	0.081	0.019	0.046	0.090	0.265	0.765	0.888	3.40
Pt	0.787	0.921	0.736	0.505	0.995	1.06	0.52	1.55	1.88	7.14	157	47.9	13.8
Pd	2.29	5.19	0.27	2.02	0.21	0.98	0.24	1.69	4.18	16.5	38.6	31.7	22.5
Pd/Ir	42.1	89.6	6.3	17.5	3.1	11.8	4.4	20.6	21.8	26.4	43.7	23.1	3.1
Cu/Pd	12815	3205	59044	9179	128146	134190	127216	19794	42630	19494	46369	88555	384378

(continued on next page)

Table 1 (continued)

Location	II-36-ZK86												
Rock type	Lh	Lh	Lh	Lh	Lh	Lh	Lh	Lh	Lh	Lh	DS-SD	DS-SD	DS-SD
Depth (m)	85	98	113	139	167	193	193	227	259	271	310	322	338
Sample	JD04-99	JD04-100	JD04-101	JD04-89	JD04-93	JD04-90	JD04-91	JD04-94	JD04-92	JD04-95	JD04-88	JD04-87	JD04-86
<i>Oxides in wt%</i>													
SiO ₂	36.2	38.2	32.7	25.2	24.9	0.8	23.1	24.4	27.3	19.0			
TiO ₂	0.24	0.29	0.08	0.18	0.13	0.01	0.16	0.11	0.11	0.26			
Al ₂ O ₃	3.17	3.51	2.49	0.84	0.72	0.03	0.77	1.47	2.45	0.81			
Fe ₂ O ₃ (t)	18.9	16.6	20.5	30.8	30.7	48.3	30.3	29.4	25.5	37.3	58.7	19.8	56.7
MnO	0.18	0.16	0.16	0.17	0.14	0.04	0.17	0.17	0.19	0.22			
MgO	30.2	30.6	30.8	26.2	25.3	0.2	25.8	25.7	25.3	21.5			
CaO	2.25	2.48	2.15	0.45	0.15	0.03	0.30	1.02	2.04	0.19			
Na ₂ O	0.47	0.51	0.29	0.03	0.02	0.01	0.01	0.18	0.34	0.01			
K ₂ O	0.23	0.26	0.11	0.07	0.03	0.01	0.02	0.09	0.14	0.01			
P ₂ O ₅		0.00		0.00	0.00					0.03			
S	3.69	2.43	5.04	10.9	11.4	25.1	10.5	10.5	8.6	14.1	24.0	4.68	25.0
LOI	7.42	5.8	7.61	9.97	10.7	27.2	10.7	9.08	6.8	3.69			
Total													
<i>Trace elements in ppm</i>													
Ni	4312	3602	12016	25588	23598	67867	21426	24883	21960	31986	54133	9868	55300
Cu	15381	6934	9190	3050	922	15352	5628	8279	10063	1792	12577	6140	13000
<i>Platinum-group elements in ppb</i>													
Ir	5.61	1.11	25.8	23.2	17.4	7.69	12.6	5.7	9.0	9.5	14.2	11.7	10.9
Ru	7.49	1.45	9.82	29.7	36.1	12.3	21.7	11.4	17.7	16.1	29.6	19.6	10.8
Rh	2.01	0.726	28.56	11.88	9.89	8.23	10.6	4.31	4.51	6.32	14.8	4.52	8.22
Pt	131	75.5	70.7	31.9	87.2	35.2	12.3	123	3.92	3.31	8.1	30.5	1.09
Pd	60.1	27.3	153	105	224	140	361	254	46.8	185	225	32.5	115
Pd/Ir	10.7	24.7	5.9	4.5	12.9	18.2	28.6	44.3	5.2	19.5	15.9	2.8	10.6
Cu/Pd	255863	254256	60119	29072	4124	109655	15609	32653	215111	9681	55899	188928	113043
Location	II-36-ZK86										Surface traverse A–A'		
Rock type	MS	MS	MS	MS	MS	MS	MS	MS	MS	MS	PLh	Lh	Lh
Depth (m)	465	467	468	468.5	469	469.5	470	470.5	471	471	4	44	64
sample	JD04-70	JD04-69	JD04-68	JD04-67	JD04-66	JD04-65	JD04-64	JD04-63	JD04-62	JD04-61	JC05-1	JC05-2	JC05-3
<i>Oxides in wt%</i>													
SiO ₂											38.1	41.9	42.4
TiO ₂											0.68	0.59	0.38
Al ₂ O ₃											13.6	7.42	7.77
Fe ₂ O ₃ (t)	59.6	60.1	60.8	58.6	59.4	63.2	56.3	56.5	56.4	65.1	10.0	12.7	12.5
MnO											0.096	0.152	0.124
MgO											20.9	24.1	25.3
CaO											10.18	4.99	5.14
Na ₂ O											1.44	1.23	1.56
K ₂ O											0.406	0.486	0.409

Line missing

P ₂ O ₅											0.064	0.085	0.052
S	25.8	23.4	22.7	27.8	25.1	26.2	26.9	23.9	26.8	23.9	0.51	0.44	0.28
LOI											3.5	5.4	3.7
Total											99.1	99.1	99.4
Trace elements in ppm													
Ni	61700	54334	58785	60100	64900	67707	58600	58600	48200	62600	201	781	793
Cu	12700	12546	10989	14200	12300	19316	13700	12200	12600	17700	80.5	82.8	50.4
Platinum-group elements in ppb													
Ir	13.2	42.0	18.6	12.2	44.1	11.1	34.4	40.7	8.56	13.6	0.17	0.030	0.055
Ru	22.4	112.0	35.3	13.7	111	22.4	94.3	113	8.05	29.6	0.390	0.190	0.500
Rh	13.5	23.1	15.7	12.3	23.5	14.0	21.9	19.4	13.2	14.9	0.07	0.01	0.045
Pt	31.0	5.5	11.7	0.81	2.2	8.0	3.0	1.6	0.51	8.2	2.42	0.85	0.645
Pd	93.4	91.9	138	48	60.1	233	51.8	102	109	123	2.39	0.80	1.51
Pd/Ir	7.1	2.2	7.4	3.9	1.4	21.0	1.5	2.5	12.7	9.0	14.1	26.7	27.4
Cu/Pd	135974	136524	79630	295833	204659	82901	264479	119608	115596	144490	33698	103528	33472
Location	Surface traverse A–A' (sampled southwards from the north boundary)									Ultramafic rocks in Noril'sk			
Rock type	Lh	Lh	Dunite	Dunite	Dunite	Dunite	Dunite	Dunite	Lh	OGD(9)	PLGD(1)	PGD(9) ^a	
Depth (m)	86	110	126	154	188	212	244	264	296				
Sample	JC05-4	JC05-5	JC05-6	JC05-7	JC05-8	JC05-9	JC05-10	JC05-11	JC05-12				
Oxides in wt%													
SiO ₂	40.6	39.0	35.5	35.7	35.0	35.9	34.7	34.4	38.2	44.5–47.0	45.1	34.8–42.6	
TiO ₂	0.38	0.27	0.17	0.16	0.14	0.15	0.14	0.15	0.26	0.6–0.88	0.7	0.41–0.66	
Al ₂ O ₃	6.45	5.37	2.60	2.58	2.50	2.70	2.23	2.31	3.77	12.5–18.8	13.9	5.7–10.1	
Fe ₂ O ₃ (t)	12.6	12.6	13.2	14.1	13.2	13.4	12.8	13.0	14.1	7.7–13.3	11.34	13.2–20.6	
MnO	0.123	0.123	0.114	0.114	0.116	0.092	0.102	0.112	0.105	0.13–0.22	0.19	0.2–0.22	
MgO	27.4	29.1	34.2	34.1	34.0	33.2	32.9	33.5	33.2	8.9–14.6	15.5	19.8–25.1	
CaO	4.40	3.10	1.05	0.96	0.73	0.63	1.11	0.58	2.01	9.1–11.5	9.3	4.4–7.2	
Na ₂ O	0.84	1.69	0.17	0.11	1.74	0.11	0.48	0.23	0.17	1.2–1.9	1.53	0.69–1.01	
K ₂ O	0.304	0.211	0.059	0.036	0.079	0.015	0.031	0.057	0.039	0.32–0.49	0.33	0.09–0.29	
P ₂ O ₅	0.050	0.038	0.030	0.031	0.031	0.035	0.031	0.030	0.041	0.07–0.11	0.08	0.05–0.10	
S	0.28	0.20	0.13	0.12	0.10	0.11	0.11	0.15	0.20	0.04–0.69	0.12	0.14–5.60	
LOI	5.7	8.0	11.6	10.6	12.0	12.9	15.2	13.9	7.8	1.0–1.7	1.0	2.4–6.8	
Total	98.8	99.5	98.7	98.4	99.5	99.1	99.7	98.3	99.6				
Trace elements in ppm													
Ni	942	968	1158	2023	1163	1141	1117	1161	1042	210–860	430	740–8700	
Cu	61.5	72.1	38.5	441.2	36.2	35.9	42.0	30.2	37.7	65–1050	122	270–13800	

Platinum-group elements in ppb

Ir	0.030	0.020	0.24	0.040	0.020	0.020	0.020	0.030	0.6–3.8	1.8	4.8–140
Ru	0.280	0.240	1.06	0.360	0.350	0.340	0.200	0.190	0.7–10.0	5.5	12–63

(continued on next page)

Table 1 (continued)

Location	Surface traverse A–A' (sampled southwards from the north boundary)										Ultramafic rocks in Noril'sk		
	Lh	Lh	Dunite	Dunite	Dunite	Dunite	Dunite	Dunite	Dunite	Lh	OGD(9)	PLGD(1)	PGD(9) ^a
Rock type	86	110	126	154	188	212	244	264	296	JC05-12			
Depth (m)	JC05-4	JC05-5	JC05-6	JC05-7	JC05-8	JC05-9	JC05-10	JC05-11	JC05-12				
Sample	0.02	0.01	0.01	0.17	0.02	0.01	0.02	0.01	0.01				
Rh	0.75	0.72	0.41	2.64	0.67	0.59	1.33	0.51	0.71		0.6–21.0	8.1	34–190
Pt	0.2	0.34	0.23	1.42	0.43	0.19	0.21	0.13	0.26		5–130	59	220–1100
Pd	25.0	24.0	20.5	11.0	16.8	29.5	66.5	25.5	23.7		3–410	46	420–3500
Pd/Ir	81993	100179	93929	167106	54082	60849	31582	59308	53045				
Cu/Pd													

Lh, lherzolite; NT-SD, net-textured sulfide dunite; DS-SD, disseminated sulfide dunite; MS, massive sulfides; PLh, plagioclase lherzolite "Depth" for the surface samples means the distance from the northern margin of the intrusion.

^a Ultramafic intrusive rocks in the sulfide-bearing intrusions in Talnakh junction: ODG: olivine gabbrodolerite; PLDG: picritic-like gabbrodolerite; PGD: picritic gabbrodolerite (Czamanske et al., 1995).

the rocks and the ores were recalculated to a volatile- and sulfide-free basis in order to investigate compositional variation in the silicate rocks. The lherzolites of Segment II contain 19–38 wt% MgO, 2.5–11.2 wt% Al₂O₃, and 1.5–6.7 wt% CaO. Their MgO contents increase from the western portion to the eastern portion, while their Al₂O₃ and CaO contents decrease. The dunites have high MgO contents (37–43 wt%) and low Al₂O₃ (0.8–3.1 wt%) and CaO (0.3–3.2 wt%). In contrast, the pyroxinites are marked by high CaO contents (10–15 wt%) and low MgO (<21 wt%).

Consistent with the concentric zonation of lithologies in the western portion of Segment II (Fig. 2), data for samples from drill hole ZK83 display a central disposition of the major elements (Fig. 3A). From the lherzolites at the upper and lower margins to the central No. 1 ore body, the MgO contents increase from 17 wt% to 43 wt%, while Al₂O₃ and CaO contents decrease from 11.2 to 0.82 and 6.7 wt% to 0.28 wt%, respectively, with the exception of a few samples (Fig. 3A). In contrast, data from bore hole ZK86 drilled in the eastern portion of the segment indicate that there is a gentle decline in MgO values, and an increase in CaO and Al₂O₃ values, on progressing through the dunite from the surface to the lherzolite containing disseminated sulfides (in the upper part of No. 2 ore body) (Fig. 3B). This is followed by a sharp increase in MgO and decreases in CaO and Al₂O₃ in the underlying net-textured sulfide-dunites (in the lower part of No. 2 ore body) (Fig. 3B). Along the surface traverse (A–A') across the eastern portion of the segment (Fig. 2), MgO decreases from 34 to 24 wt%, Fe₂O₃(t) decreases from 14 to 12.5 wt% from the dunites to the underlying lherzolite, whereas, Al₂O₃ and CaO increase from 2.2 to 7.8 wt% and 0.6 to 5 wt% respectively (Fig. 4A and B).

4.2. Nickel, copper, and platinum-group elements

There is a central disposition of Ni, Cu, and S in whole rock samples from drill hole ZK83, with concentrations of these elements significantly higher in the sulfide-dunites in No. 1 ore body than in the lherzolites at the margins (Fig. 3C). The Ni and S contents of the sulfide-dunites are quite uniform whereas Cu contents are much more variable. The PGE in the ZK83 are also centrally disposed (Fig. 5A). With one exception, the Pt and Pd contents of the lherzolites in the drill hole range from 1.4 to 3.7 ppb and 1.7 to 6.0 ppb, respectively. The Ir contents of the lherzolites range from 0.06 to 0.42 ppb (Fig. 5A). By contrast, the dunites hosting both net-textured and disseminated sulfides in the No. 1 ore body are strongly enriched in Pd, Pt and Ir. Platinum varies from 14.6 to 279 ppb, Pd from 62.1 to 525 ppb and Ir from 4.8 to 116 ppb (Fig. 5A). There is a general increase in Pt contents and decrease in Ir with depth in the sulfide-dunites, whereas Pd has a more irregular distribution pattern (Fig. 5A).

For samples from the eastern Segment II (drill hole ZK86), there is a marked general increase in Cu, Ni, and S contents from the dunite and lherzolite through to the sulfide-dunites and the massive sulfides that comprise No. 2 ore body (Fig. 3D). The Cu contents of the lherzolites range

Table 2

Blank (ng), detection limits (DL) (ng/g) and analytical results (ng/g) of reference materials, UMT-1 and WPR-1.

	Blank	UMT-1 (Ultramafic Ore Tailings)		WPR-1 (Peridotite)	
		Certified	Qi et al. (2004) (Mean $\pm$ S)	Certified	Qi and Zhou (2008) (Mean $\pm$ S)
Ir	<0.15	8.8 $\pm$ 0.6	8.55 $\pm$ 0.65	13.5	13.8 $\pm$ 1.2
Ru	<0.15	10.9 $\pm$ 1.5	9.53 $\pm$ 0.82	22	23.1 $\pm$ 1.9
Rh	<0.05	9.5 $\pm$ 1.1	9.22 $\pm$ 0.71	13.4	12.8 $\pm$ 0.7
Pt	<0.5	129 $\pm$ 5	123 $\pm$ 8	285	280 $\pm$ 13
Pd	<0.5	106 $\pm$ 3	115 $\pm$ 7	238	238 $\pm$ 17

S, standard deviation; UMT-1 (Ultramafic Ore Tailings); WPR-1 (Peridotite).

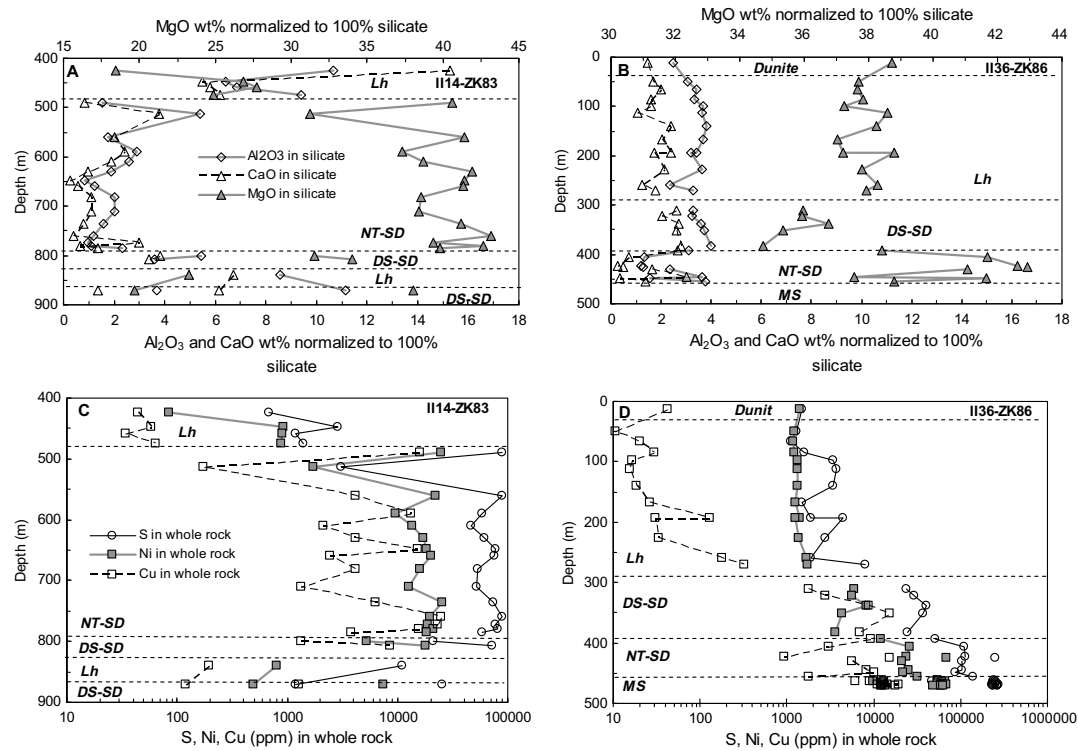


Fig. 3. Chemostratigraphic columns of the sampled bore holes. (A) and (B) The variations of Al_2O_3 , CaO , and MgO normalized to 100% silicate in the bore holes ZK83 and ZK86 in the western and eastern portions of Segment II, respectively; (C) and (D) The variations of Ni, Cu, and S, respectively. Lh, Lherzolite; NT-SD, net-textured sulfide dunite; DS-SD, disseminated sulfide dunite; MS, massive sulfides.

from 10.6 to 322 ppm, their S contents vary from 1100 to 8000 ppm, and their Ni contents have a remarkably narrow range (1100–1716 ppm) (Fig. 3D). High Cu, S, and Ni values at a depth of 425 m in ZK86 are caused by the presence of a massive sulfide vein which cross-cuts the net-textured sulfides (Table 1). Iridium and Pt values increase from 0.04 ppb and 0.37 in the uppermost dunite to 0.6 and 7.1 ppb in the lowermost lherzolites, respectively. The variation of Pd with depth is not as regular as that of Pt or Ir, with values ranging from 0.3 to 16.5 ppb Pd (Fig. 5B). Whereas Ir and Pd generally continue to increase through the disseminated and net-textured sulfides, Pt values are highly variable. The massive sulfides have markedly lower Pt contents than the overlying disseminated and net-textured sulfides, whereas the Ir and Pd contents of the massive sulfides are similar to those of the overlying net-textured sulfides (Fig. 5B).

When normalized to 100% sulfide (see appendix), the disseminated ores (with 1.4–3.7 wt% S) and net-textured

ores (with 3.7–15 wt% S) in the No. 1 and No. 2 ore bodies have similar Ni (7–16 wt%) but large variations in Cu (0.3–15 wt%) contents and exhibit a large range in Ni/Cu ratios (Fig. 6A). The massive sulfides (with >15 wt% S) not only have the smallest variation in Cu tenor of the three ore types on a silicate-free basis, but also the narrowest range in Ni/Cu ratios (Fig. 6A). The massive sulfides have lower $\text{Pd}_{\text{sulfide}}$ (subscript “sulfide” means normalized to 100% sulfide) and much lower $\text{Pt}_{\text{sulfide}}$ than the disseminated sulfide and the net-textured sulfides (Fig. 6B). The massive sulfides are also marked by very low Pt/Pd ratios (0.1–0.01), which are much lower than those of the disseminated and net-textured sulfide ores ($\text{Pt}/\text{Pd} > 0.1$) (Fig. 6B). In addition, the massive sulfides are also generally lower in $\text{Rh}_{\text{sulfide}}$ and $\text{Ir}_{\text{sulfide}}$ than the net-textured and disseminated sulfides, although they plot on the same well defined trend (Fig. 6C). The differences between the massive sulfides and the disseminated and net-textured sulfides are shown in the diagram of Ir against Pt (Fig. 6D).

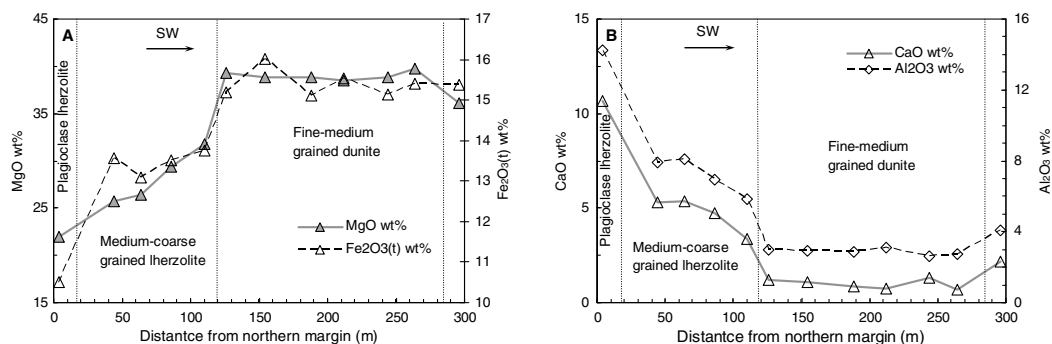


Fig. 4. Chemical variations of surface traverse A-A' across Segment II of the Jinchuan intrusion as shown in Fig. 2A. (A) Variations of MgO and Fe₂O₃(t) (total iron oxide); (B) variations of CaO and Al₂O₃.

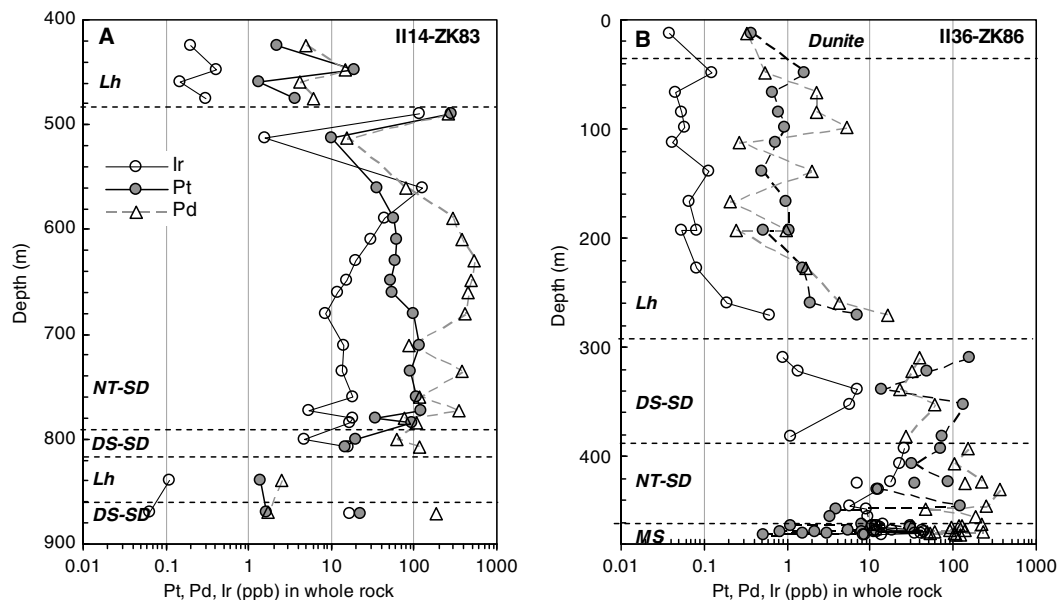


Fig. 5. Variation of Pt, Pd, and Ir in the bore holes ZK83 (A) and ZK86 (B) which intersected the western and eastern portions of Segment II, respectively. Lh, lherzolite; NT-SD, net-textured sulfide dunite; DS-SD, disseminated sulfide dunite; MS, massive sulfides.

The sulfides of the No. 1 ore body have higher PGE concentrations in 100% sulfide than those of No. 2 ore body. The disseminated and net-textured ores of No. 1 ore body contain 90–1600 ppb Pt_{sulfide}, 400–9800 ppb Pd_{sulfide} (up to 10 ppm in one sample), 40–400 ppb Rh_{sulfide}, and 30–800 ppb Ir_{sulfide} (Fig. 6B–D). The disseminated and net-textured sulfides of No. 2 ore body have wide variations in Pt_{sulfide} (9–3200 ppb), and are low in Pd_{sulfide} (240–1500 ppb), Rh_{sulfide} (15–280) ppb, and Ir_{sulfide} (17–250 ppb) (Fig. 6B–D).

5. DISCUSSION

5.1. Comparison with the Noril'sk-Talnakh and Voisey's Bay deposits

The Jinchuan deposit has some features that are different from other magmatic sulfide deposits formed in magma conduits, such as the Noril'sk-Talnakh and Voisey's Bay deposits. In the Noril'sk-Talnakh region, Russia, the massive sulfides are concentrated at the bases of the ore-bearing

sills and overlain by picritic gabbrodolerites with disseminated sulfides, which in turn are overlain by gabbrodolerites (Naldrett et al., 1992; Czamanske et al., 1995). Many authors (Brugmann et al., 1993; Ripley et al., 2003; Lightfoot and Keays, 2005; Naldrett et al., 1995) have suggested that these ore-bearing sills were the magma conduits for the overlying basalts. However, Arndt et al. (2003) and Latypov (2007) argue that the magmas that formed the sills are not represented in the overlying basalts. Brugmann et al. (1993) and Lightfoot and Keays (2005) have proposed that the sulfides in the sills were formed in a deep-seated staging chamber and carried into the sills by more primitive magmas than the crustal contaminated magmas that were the source of the Ni, Cu, and PGE in the ores.

The Voisey's Bay Ni–Cu–Co sulfide deposit consists of a series of chambers that comprise a magma conduit system (Naldrett et al., 2000). The sulfides were formed in a staging chamber at depth by segregation of magmatic sulfides from a crustally contaminated magma and then transported by a second magma into the overlying conduit where they settled at the bases of the small chambers along the conduit (Li and

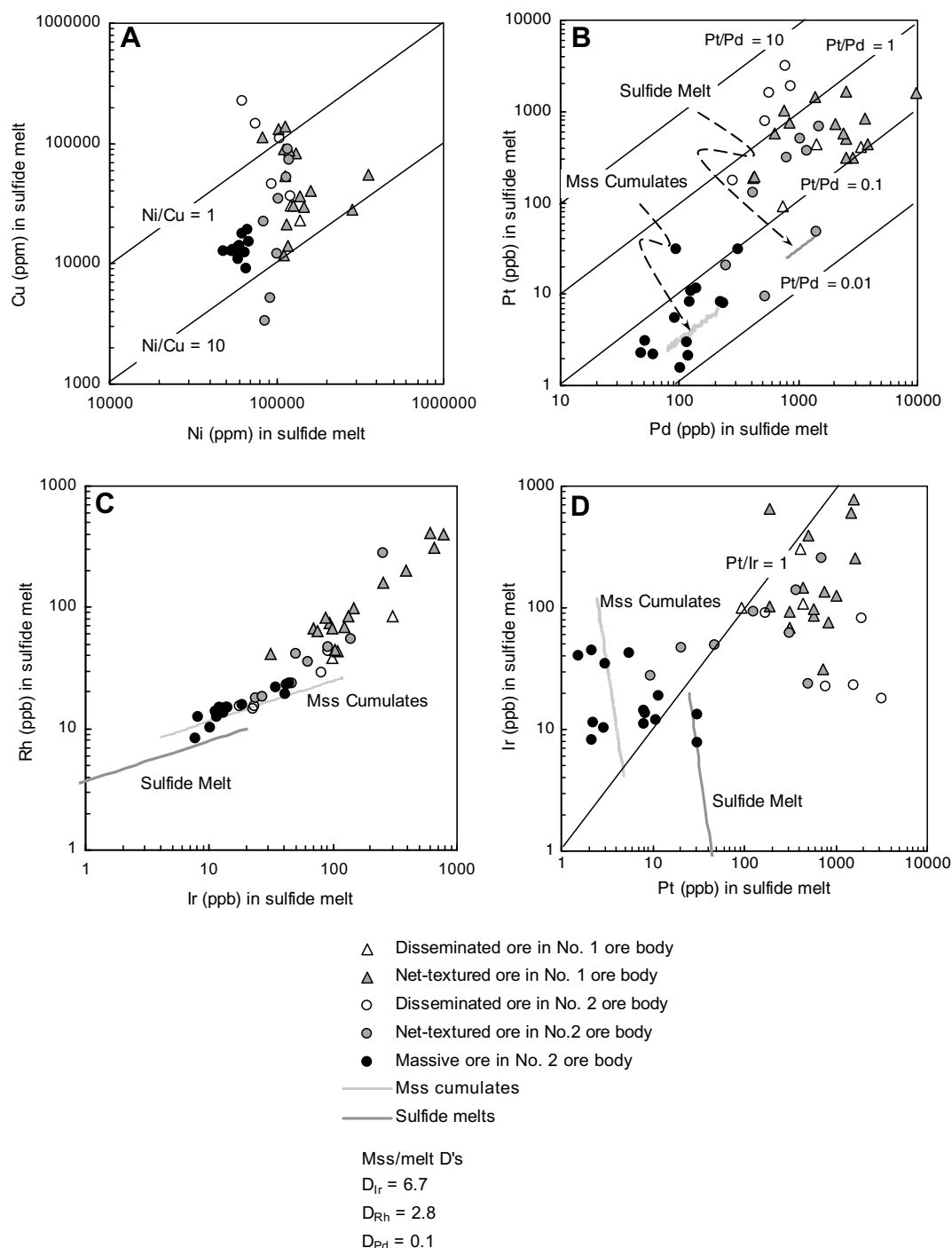


Fig. 6. Variation of Ni, Cu, and PGE in the sulfide ores on a 100% sulfide basis. (A) Scattergram showing that the Jinchuan ores have a wide range in Cu contents but a relatively narrow range in Ni contents. (B) Scattergram showing that Pt generally increases with Pd contents and that the massive ores generally have low to very low Pt/Pd ratios. (C) Scattergram showing the strong correlation between Ir and Rh, and (D) Scattergram between Ir and Pt that demonstrates that the massive sulfides have much lower Pt contents for comparable Ir contents than the other ore types. “Mss Cumulates” and “Sulfide Melt” shown in (B, C, and D) are variations of Pt, Pd, Ir, and Rh in the Mss cumulates and the residual sulfide melts due to up to 50% fractional crystallization of the sulfide liquids.

Naldrett, 1999). For the ore-bearing intrusions in both the Noril'sk-Talnakh and Voisey's Bay areas, the Ni, Cu, PGE, and MgO contents of the rocks and Fo number of olivines increase downwards, indicating an *in-situ* fractionation (Czamanske et al., 1995; Li and Naldrett, 1999).

In contrast, rocks in the western portion of Segment II that host the No. 1 ore body are concentrically zoned, with sulfide-dunite enveloped by ilmenite (Fig. 2); the oxides, Ni, Cu, and PGE are also concentrically zoned in this portion of the Jinchuan intrusion (Figs. 3A, C and 5A). In the

eastern portion of Segment II, although Cu and PGE increase towards its base (Figs. 3D, 5B), MgO decreases from the dunite at the top downwards to the lherzolite (Fig. 3B). In the middle to east portion of Segment II, the lherzolite is overlain by dunite (Figs. 2 and 4). On the other hand, the olivine crystals of the dunites and lherzolites of the Jinchuan intrusion have a very narrow range of Fo (81.5–86.5) (Li et al., 2004). **These features have led some authors to propose that the Jinchuan intrusion was formed by injection of sulfide-free and sulfide-bearing olivine crystal mushes from a deep-seated staging magma chamber, where crystallization of olivine and sulfide segregation occurred (De Waal et al., 2004; Li et al., 2004; Song et al., 2006).**

The Jinchuan disseminated and net-textured sulfides have much lower PGE contents on a 100% sulfide basis than the Noril'sk-Talnakh sulfides, which contain 2–40 ppm Pt_{sulfide}, 8–71 ppm Pd_{sulfide}, and 8–750 ppb Ir_{sulfide} (Naldrett, 2005), and higher than that of the sulfide ores of the Leopard Troctolite in Voisey's Bay, which contain 34–500 ppb Pt_{sulfide}, 100–520 ppb Pd_{sulfide}, and 2–9 ppb Ir_{sulfide} (Naldrett et al., 2000) (Fig. 7). In addition, the massive sulfide and the overlying net-textured sulfides in the No. 2 ore body show strongly negative Pt-anomalies (Fig. 7).

5.2. Prior sulfide segregation

Continental flood basalts formed by moderate to high degrees of partial melting that leave no sulfides in the mantle source, such as the Siberian Traps and the Emeishan flood basalts, are generally high in Pt and Pd (~10–20 ppb) and have moderate Ir contents (Lightfoot and Keays, 2005; Bockrath et al., 2004; Arndt et al., 2005; Song et al., 2006). However, continental flood basalts may become PGE-depleted if they become S-saturated due to crustal contamination (Brugmann et al., 1993; Lightfoot and Keays, 2005; Song et al., 2006, 2008). Previous studies indicate that the Jinchuan intrusive rocks are enriched in the LREE, similar to continental flood basalts (Song et al., 2006; Lehmann et al., 2007). Calculations based on the composition of the olivine and the whole rocks indicate that the parental magma of the Jinchuan intrusion was a high-Mg basaltic magma (11.5 wt% MgO, 1.3 wt% Na₂O, and 0.8 wt% K₂O) that was related to continental rifting (Chai and Naldrett, 1992b; Lehmann et al., 2007). Thus, we assume that the primary magma of the Jinchuan intrusion was originally high in Pt and Pd and moderate in Ir.

The Ir and Pd contents of most Jinchuan dunites and lherzolites are significantly lower than those of the average Siliceous High Magnesium Basalts (SHMB) of the Archaean Yilgarn Craton, Western Australia (Fig. 8; Redman and Keays, 1985). The magmas that formed the SHMB contain ~12 wt% MgO (cf. Hamlyn and Keays, 1986) and hence had similar MgO contents to that of the Jinchuan magma (11.5 wt% MgO, Chai and Naldrett, 1992b). The significantly low Ir and Pd contents of the Jinchuan rocks suggest a loss of PGE due to prior sulfide separation (Fig. 8).

Prior sulfide segregation can also be estimated by Cu/Pd ratios of the sulfides and the residual magmas. The concentration of a metal in a sulfide melt segregated from a silicate melt can be described by following equation (Campbell and Naldrett, 1979):

$$Y_i = X_{i0}^* D_i^* (R + 1) / (R + D_i) \quad (1)$$

where Y_i and X_{i0} represent the concentrations of element i in the sulfide melt and silicate liquid, respectively, D_i is sulfide/silicate liquid partition coefficient of i , and R is the ratio of the volume of the silicate melt to that of the segregated sulfide melt. According to Eq. (1), the relationship between Cu/Pd ratios in a sulfide melt and the silicate melt with which it is in equilibrium is (Campbell and Barnes, 1984):

$$X_{Cu0}/X_{Pd0} = Y_{Cu}/Y_{Pd} D_{Pd}/D_{Cu}^* (R + D_{Cu}) / (R + D_{Pd}) \quad (2)$$

In our calculations, D_{Pt} and $D_{Ir} = 3 \times 10^4$, $D_{Pd} = 4 \times 10^4$, D_{Ni} and D_{Cu} are 400 and 1000, respectively (Francis, 1990; Fleet and Stone, 1991; Peach et al., 1994; Bezmen et al., 1994). It is assumed that the partition coefficients remained constant during sulfide segregation, although the values are dependent on f_{O2} , f_{S2} , and the compositions of the silicate and sulfide liquids (Rajamani and Naldrett, 1978; Peach et al., 1990; Stone et al., 1990). **The calculations indicate that the Cu/Pd values of the magma in equilibrium with the Jinchuan sulfides (Cu/Pd ~ 10^4 – 10^5) (Table 1) were as large as 2×10^4 to 9×10^6 under R factors of 10^3 – 10^4 . Such Cu/Pd ratios are much higher than those of the primitive mantle (Cu/Pd ~ 10^4) (Barnes and Maier, 1999). These data also suggest that the Jinchuan sulfides were segregated from magmas that had experienced a prior S-saturation event that depleted the precursor magma in Pd but not Cu.**

If the high Mg tholeiitic magma of the Jinchuan intrusion (Chai and Naldrett, 1992b) was compositionally similar to that which produced the Mokulavsky basalts of the Siberian Trap at Noril'sk, it would have contained 6.3 ppb Pt, 6.2 ppb Pd, and 0.1 ppb Ir; Naldrett et al. (1995) have suggested that the Mokulavsky magma was the magma that had transported the magmatic sulfides formed in a deep-seated staging chamber to their current sites. The Jinchuan magma contained relatively high Ni (~300 ppm) and low Cu (~62 ppm) because the Jinchuan magmas only experienced limited fractional crystallization of olivine before sulfide segregation (Li et al., 2004). Calculations using Eq. (1) indicate that when such a tholeiitic magma reached S-saturation, the segregated sulfide melt would have contained 6–60 ppm Pd and Pt and 100–750 ppb Ir if R factors were 10^3 to 10^4 , which are comparable to the Pd, Pt and Ir contents of the sulfides in the Noril'sk-Talnakh deposits (Naldrett, 2005); however, the Jinchuan sulfides contain much lower PGE contents than these calculated values (Fig. 9). The low PGE contents of the Jinchuan sulfides cannot be explained by low R factors; for example, if the R factor was as small as 100, the segregated sulfide melt should have had Ni contents much lower than that in the Jinchuan sulfides (Fig. 9). These calculations again indicate that the Jinchuan sulfides segregated from a PGE-depleted magma

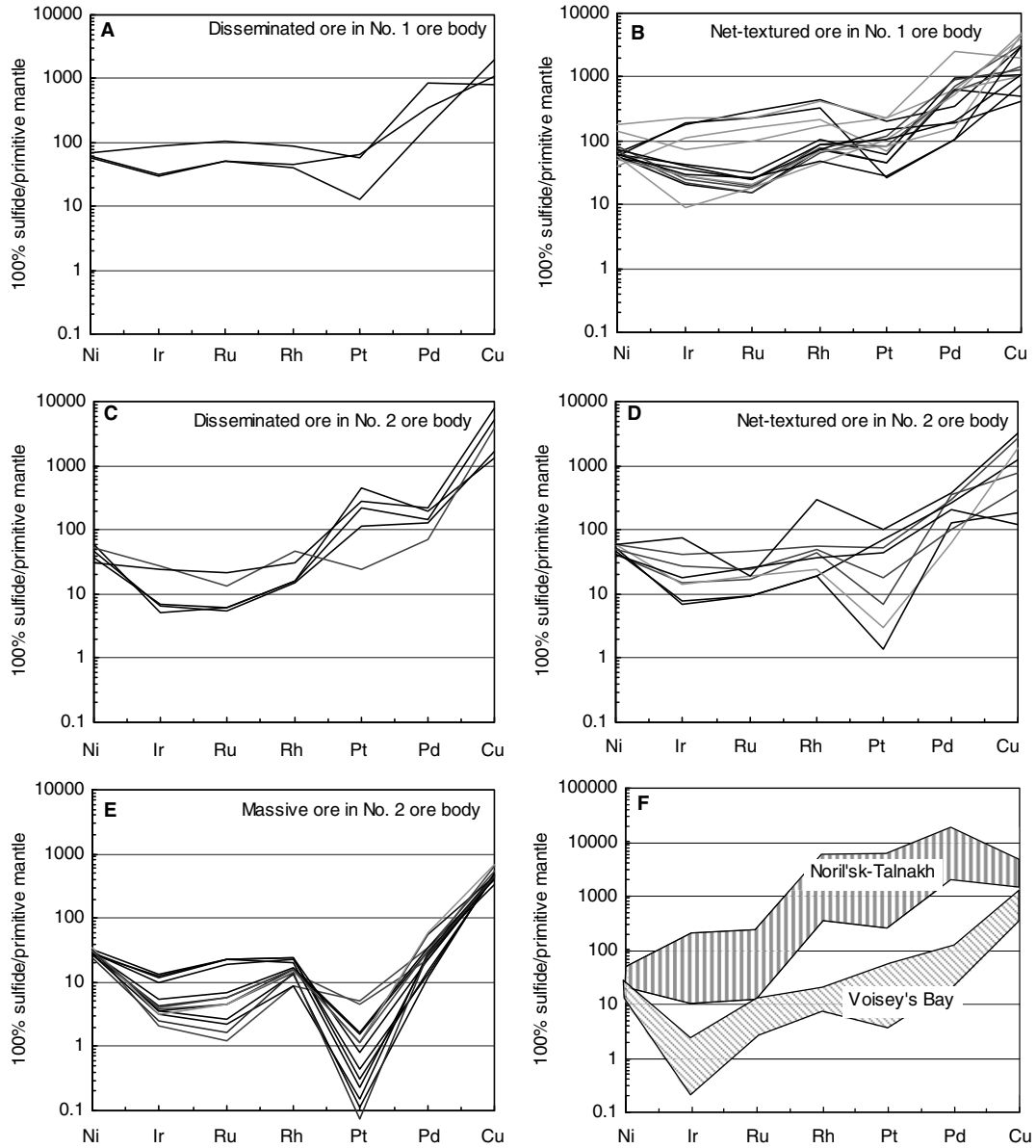


Fig. 7. Primitive-mantle normalized PGE patterns of the sulfide ores of the No. 1 and No. 2 ore bodies on a 100% sulfide basis. Primitive mantle values are from Barnes and Maier (1999). The data of the sulfide ores on a 100% sulfide basis of the Noril'sk-Talnakh deposits are after Naldrett (2005), and those of the Voisey's Bay deposit are after Naldrett et al. (2000).

that had been depleted in the PGE due to an earlier S-saturation event, as proposed by Ihlenfeld and Keays (2006).

5.3. Quantitative approach to sulfide segregation

The concentration of an element in the residual melt of a fractionating magma can be determined from the Rayleigh equation:

$$C_i^L = C_i^O * F^{(D_i-1)} \quad (3)$$

where C_i^O and C_i^L are the concentrations of element i in the original silicate magma and residual liquid, respectively, after a fraction F of the melt has crystallized; D_i is the bulk partition coefficient of element i . If sulfides constitute a part of the crystallizing solids, the bulk D_i for the PGE will be

dominated by their sulfide/silicate liquid partition coefficients. If as little as 0.008% sulfide melts were separated from a high Mg basaltic magma with PGE concentrations similar to the Mokulaevsky basalts in the Noril'sk area (Naldrett et al., 1995) and ~300 ppm Ni in a deep-seated staging chamber, the residual magmas would be depleted in PGE (0.57 ppb Pt, 0.25 ppb Pd, 0.009 ppb Ir), but undepleted in Ni (290 ppm). We suggest that the Jinchuan sulfides were segregated from such a PGE-depleted magma.

Our calculations indicate that sulfide melts separated from such a PGE depleted magma under R factors between 10^3 and 10^4 could roughly match the $Pd_{sulfide}$ and $Ir_{sulfide}$ of the Jinchuan disseminated and net-textured sulfides (Fig. 9A and B). However, all the massive ores and some of the net-textured sulfides of No. 2 ore body have Pt con-

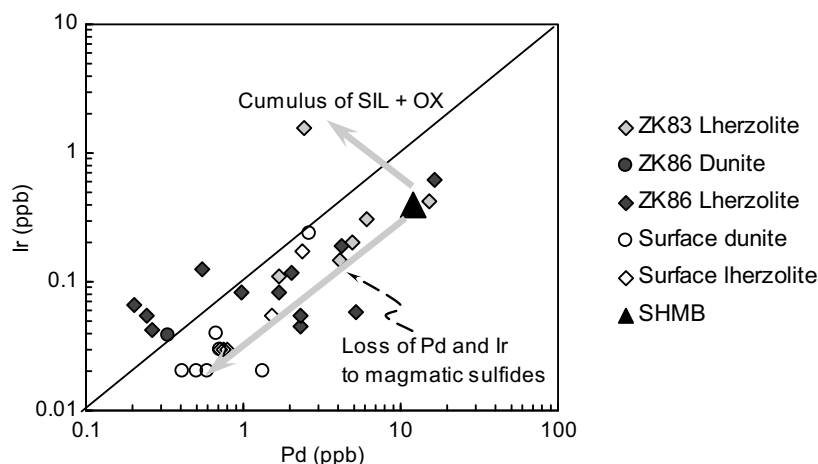


Fig. 8. Plot of Ir vs. Pd, showing that the Jinchuan sulfide-barren rocks are strongly PGE-depleted. SHMB is the average Siliceous High Magnesium Basalts of the Archaean Yilgarn Craton, Western Australia (Redman and Keays, 1985). SIL, silicate minerals; OX, oxides.

tents much lower than the secondary sulfide segregation trend (Fig. 9C). The negative Pt-anomaly of the massive ores and some of the overlying net-textured ores relative to the other PGEs is also clearly displayed in the primitive-mantle normalized PGE patterns (Fig. 7D and E). Some other mechanism(s) must therefore be responsible for the PGE compositions of the massive sulfides in the Jinchuan deposit.

5.4. Negative Pt-anomaly in the massive ores

The massive sulfide ores in Jinchuan are generally poorer in PGE than the other types of sulfide ores because they formed under lower R factors (Fig. 9); however, this would not explain their significantly negative Pt-anomalies. Although the sulfide/silicate melt partition coefficients of Pt may be about a third of that of Pd (Vogel and Keays, 1997), the difference in partition coefficients of Pd and Pt would not be sufficient to cause the observed low Pt/Pd ratios of the massive sulfides (Fig. 6B).

A possible explanation for the negative Pt-anomalies in the massive and some of the net-textured sulfides is that the sulfide melts that produced them had undergone fractionation. The earliest phase to crystallize from a sulfide melt is Fe-rich monosulfide solid-solution (Mss; Kullerød et al., 1969). Osmium, Ir, Ru, and Rh are compatible with Mss, and hence enriched in the cumulate Mss, whereas Cu, Pt and Pd are enriched in the residual sulfide melts (Keays and Crocket, 1979; Naldrett et al., 1982; Li et al., 1992; Fleet et al., 1993; Ballhaus et al., 2001; Ebel and Naldrett, 1996). The first Mss produced from the Jinchuan sulfide melt would have had relatively high Ir and Rh contents, but those that formed later would have had lower Rh and much lower Ir contents. The reason for this is that the Mss/sulfide melt partition coefficient of Ir ($D_{Ir} = 6.7$) is more than twice that of Rh ($D_{Rh} = 2.8$; Ballhaus et al., 2001). Calculations using these partition coefficients and initial concentrations of 80 ppb $Ir_{sulfide}$ and 34 ppb $Rh_{sulfide}$ indicate that many of the massive sulfides and some of the net-textured sulfides plot along the Mss cumulate trend (Fig. 6C). Hence, a mod-

el involving fractionation of a sulfide melt is permissive in accounting for the Ir and Rh contents of these particular samples. However, the $Ir_{sulfide}$ and $Rh_{sulfide}$ contents of all of the net-textured and disseminated sulfides do not lie along either the sulfide melt or Mss control lines (Fig. 6C). The case for fractionation of a sulfide melt cannot explain the negative Pt-anomaly of the Jinchuan massive ores either when one considers the $Ir_{sulfide}$ and the $Pt_{sulfide}$ data (Fig. 6D). We have used Mss/sulfide melt partition coefficients $D_{Ir} = 6.7$ and $D_{Pt} = 0.1$ (Ballhaus et al., 2001) and initial concentrations of 2600 ppb Pt and 80 ppb Ir in the sulfide melt to model the fractionation paths of the sulfide melt and Mss cumulates. None of the Ir/Pt ratios lie along either the sulfide melt or Mss cumulate fractionation paths (Fig. 6D). Finally, as $D_{Pt} = D_{Pd}$, fractionation of a sulfide melt cannot explain the much more rapid decrease in Pt relative to Pd in the Jinchuan sulfides (Fig. 6B). Hence, fractionation of a sulfide melt is not responsible for the negative Pt-anomaly of the massive and net-textured ores. The explanation for the very low Pt contents of the Jinchuan ores must lie in some other quarter.

Another possibility for the very low Pt contents of the massive ores and some of the net-textured ores is that the magmas that produced them had undergone fractionation prior to the ore-forming event. Momme et al. (2002) showed that the bulk (silicate $\pm$ oxides/silicate melt) partition coefficient of Pt, D_{Pt} , in S-undersaturated continental flood basalt magmas in East Greenland was ~ 1.0 , whereas $D_{Pd} = \sim 0.4$. Lightfoot and Keays (2005) used these bulk partition coefficients to show that the magmas which had formed the Noril'sk-Talnakh deposits had undergone extensive fractional crystallization prior to the formation of the magmatic sulfides. They inferred that as a result of silicate $\pm$ oxide fractionation, the Pt/Pd ratios of the magmas decreased from mantle values of ~ 1.8 to values of 0.4 in the most fractionated lavas; as a consequence, the Noril'sk-Talnakh Ni-Cu-PGE sulfide ores have Pt/Pd ratios of 0.29 to 0.18 (Lightfoot and Keays, 2005). However, the Pt/Pd ratios of the Jinchuan massive sulfides range from 0.1 to 0.01 (Fig. 6B). This suggests that some sort of Pt

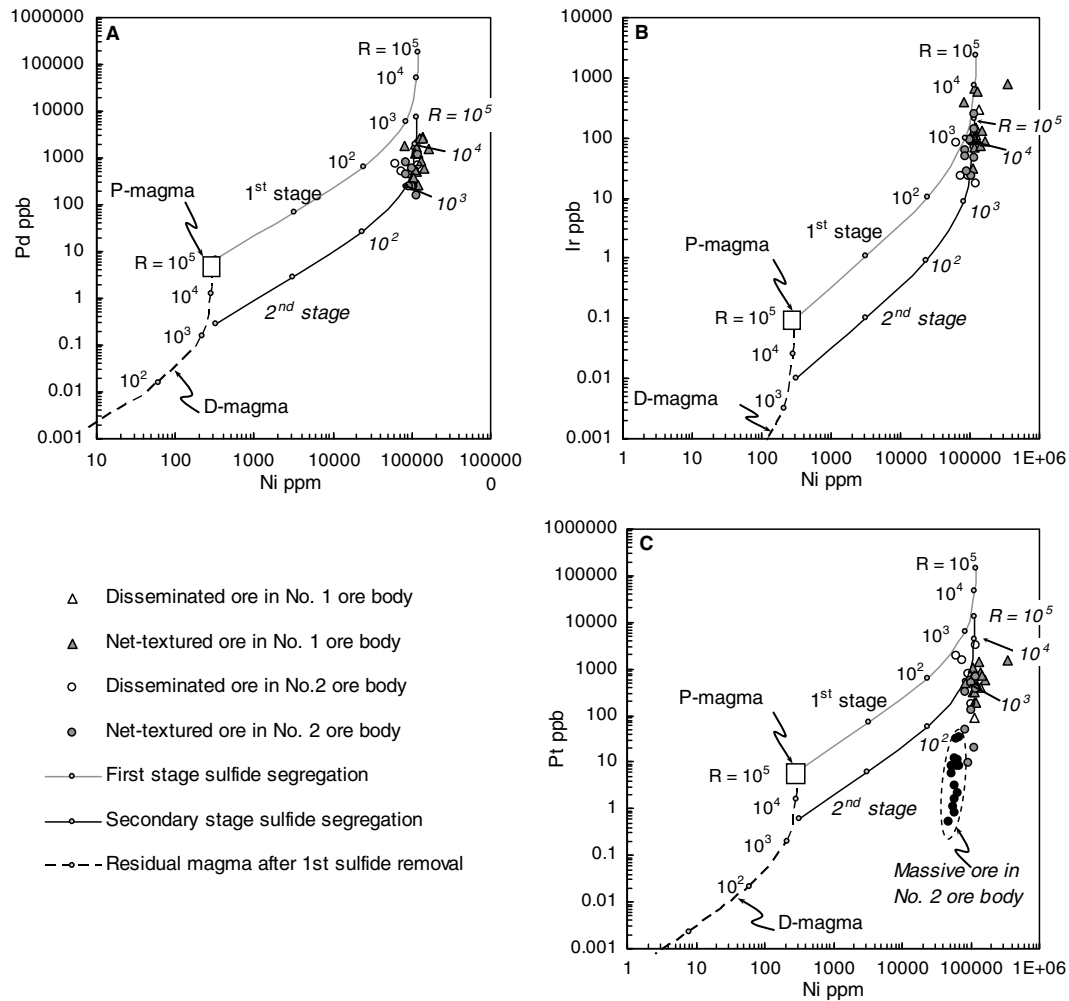


Fig. 9. Model calculations of the Ni and PGE concentrations in 100% sulfides resulting from sulfide segregation from PGE undepleted and PGE depleted magmas under various R factors. First stage sulfide segregation from high Mg basaltic magma with 6.3 ppb Pt, 6.2 ppb Pd, 0.1 ppb Ir, 300 ppm Ni, and 62 ppm Cu. Secondary stage sulfide segregation from PGE-depleted magma with 0.57 ppb Pt, 0.25 ppb Pd, 0.009 ppb Ir, 290 ppm Ni, and 52 ppm Cu. The contents of PGE in the sulfide liquids separated during the first and second stages sulfide segregation are shown by the lines labeled as 1st stage and 2nd stage under variable R-factors (labeled as $R = 10^i$). P-magma and D-magma mean the PGE-undepleted original magma and the residual magma moderately depleted in PGE after the first stage sulfide segregation.

bearing alloy had co-precipitated with the silicates $\pm$ oxides during fractionation of the magma batch that formed the massive sulfides.

Iron–platinum alloy grains are included in chromite in concentrically zoned mafic–ultramafic intrusions of the Urals and Alaska (Nixon and Hammack, 1991; Cabri, 1992) and the Heazlewood Complex, Tasmania (Peck and Keays, 1990a,b). These observations suggest that the Fe–Pt alloys precipitated directly from the magma very early as liquidus phases together with chromite. The Fe–Pt alloys are extremely rich in Pt and sometimes IPGE as well (Nixon and Hammack, 1991; Peck and Keays, 1990a,b). Chromite is one of the earliest crystallized phases in the Jinchuan intrusion and always armoured by olivine (Barnes and Tang, 1999). A possible interpretation of the negative Pt-anomaly in the Jinchuan sulfides is that the high-Mg basaltic magmas which produced the massive sulfides experienced separation of Fe–

Pt alloys during their early stages of fractional crystallization of silicates $\pm$ oxides prior to the first stage sulfide separation. The separation of Fe–Pt alloys prior to the first stage of sulfide segregation may have accounted for the Pt-depletion in the Jinchuan magmas. However, there is no obvious explanation as to why it was only the magma batch from which the massive sulfides formed that experienced prior such an extensive loss of Pt.

The massive sulfides occur as veins and hence have all been remobilized from their original sites in the Jinchuan intrusion. The low Pd/Ir ratios (<41 , Table 1) of the Jinchuan massive and net-textured sulfides indicate that the ores are magmatic in origin, because hydrothermal deposits typically have very high Pd/Ir ratios (Keays et al., 1982; Keays, 1995). The mantle-like $\delta^{34}\text{S}$ values of the sulfide assemblages at Jinchuan (mostly between -2 and 2 per mil) and the low S contents in the country rocks (<100 ppm) also suggest that the Jinchuan sulfides are

magmatic in origin (Ripley et al., 2005). It is possible that during their remobilization, Pt was selectively leached from the sulfides and deposited elsewhere in the Jinchuan system. Hydrothermal fluids can mobilize Pt without the other PGE as in the case of the hortonolite pipes of the Bushveld Complex (Stumpfl and Rucklidge, 1982; Rowell and Edgar, 1985) or Pd and Pt as in the case of the “hydrothermal” Cu–Ni sulfide stringers in the footwall basalts at Kambalda (Leshner and Keays, 1984, 2002; Leshner and Stone, 1996).

5.5. Model for the Jinchuan intrusion and No. 1 and No. 2 ore bodies

The above discussions lead us to propose that the magmatic Ni–Cu–(PGE) sulfides in the Jinchuan deposits were formed in the magma conduit system. Prior to entering the staging chamber, the S-undersaturated and PGE-undepleted high MgO basaltic magmas which formed the sulfides underwent about 0.008 wt% sulfide removal and became significantly PGE-depleted. The earliest pulses of magmas may have experienced fractionation of olivine and as well as Fe–Pt alloy. When these S-saturated and PGE-depleted basaltic magmas entered the deep-seated magma chamber, extensive sulfide segregation together with crystallization of olivine occurred as a result of crustal contamination and further fractional crystallization (Li et al., 2004; Song et al., 2006). The sulfides were PGE-poor, and additionally the sulfides separated from the earliest pulses of magma were either PGE-poor or Pt-depleted (Fig. 9C). The large variations of $Pd_{sulfide}$ and Cu/Pd values of the No.1 and No. 2 ore bodies (Fig. 10) suggest that the staging chamber was an open system and many pulses of magma with variable PGE concentrations were involved in the process. Magma replenishment provided more and more material for further sulfide segregation and a relatively constant compositional environment for crystallization of olivine with a narrow range of Fo numbers (Li et al., 2004). The new pulses of magmas entered the staging chamber along the base of the sulfide- and crystal-free layer because of their moderate density. The sulfide- and crystal-free magmas at the upper layer of the chamber were pushed out to form lava flows or sulfide-poor intrusions because of replenishment of fresh magma. In this model, the staging magma chamber was an open system magma conduit (Fig. 11).

Sulfides separated from the early pulses of PGE-depleted magma settled to the base of the staging chamber, whereas those that formed later only settled part way down through the olivine crystal mushes formed by earlier pulses of magma (Fig. 11). The later sulfide melts did not mixed completely with the sulfides that had settled earlier and homogenize with them because of obstruction by the cumulate olivine grains. It is possible that the sulfide melts that accumulated at the base of the chamber formed a pure sulfide melt layer and that these had low PGE contents because they had formed under low R factor conditions; in addition, the magma batches that formed the massive sulfides may have experienced loss of Fe–Pt alloys prior to becoming S-saturated.

Following segregation of the sulfides, normal fault activity compressed the staging chamber and squeezed the upper sulfide-poor olivine crystal “mushes” followed by the sulfide-bearing olivine crystal “mushes” upwards into the Jinchuan intrusion (Fig. 11). The sulfide-poor olivine crystal “mushes” are now represented by the sulfide-barren ultramafic rocks in the Jinchuan intrusion (Fig. 2). The sulfide-bearing olivine-mushes in the lower layer of the staging chamber were emplaced into the western part of Segment II where they formed the No.1 ore body (Figs. 2 and 11). Flow differentiation created the concentrically zoned composition distribution (Li et al., 2004). The sulfide-bearing olivine-mushes in the lower levels of the staging chamber were also emplaced into the eastern part of Segment II where they formed the No. 2 ore body.

The massive sulfide ores may have been formed via two different mechanisms. It is possible that they were formed from a pure sulfide layer that had accumulated at the base of the staging chamber; in this scenario, the magma batches that formed these sulfides had been depleted in Pt due to the segregation of Fe–Pt alloys prior to the magmas becoming S-saturated (Figs. 2 and 11). Another possibility is that the massive sulfides were remobilized from net-textured sulfides after emplacement of the olivine-sulfide mushes into Jinchuan chamber. During remobilization of the sulfides, Pt was selectively leached by hydrothermal fluids from them and deposited elsewhere in the Jinchuan system.

5.6. Significances of Jinchuan style ore system

The Jinchuan deposit has several unique features and can be termed as the Jinchuan style ore system. The most important feature is that the giant ore bodies are hosted in a relatively small ultramafic intrusion and the sulfides have low PGE contents in 100% sulfide basis. Another significant feature is that the sulfide-free rocks in the middle portion of the Jinchuan intrusion are strongly PGE-depleted. These unique features of the Jinchuan style ore system not only are significant for understanding the formation of sulfide ore bodies in a magma conduit system, but also is exploration perspective.

Our model for the Jinchuan intrusion and No. 1 and No. 2 ore bodies proposes that the PGE-depleted ultramafic rocks and sulfide ore bodies in the Jinchuan intrusion were formed by injections of sulfide-free and sulfide-bearing olivine crystal mushes, respectively, from a deep-seated staging magma chamber into their current positions. Similar processes are not known in the other magmatic sulfide deposits formed in magma conduit system, such as the Noril'sk–Talnakh Ni–Cu–PGE deposit and the Voisey's Bay Ni–Cu–Co deposit (Li and Naldrett, 1999; Naldrett et al., 1992, 2000; Czamanske et al., 1995; Naldrett and Lightfoot, 1999; Arndt et al., 2003; Lightfoot and Keays, 2005), although the sulfide segregation process occurred in the deep-seated staging chamber is similar in some respects to that in the sills that were feeders to the overlying flood basalts in the Noril'sk region. The low PGE tenors of the Jinchuan sulfides indicate that they were segregated from

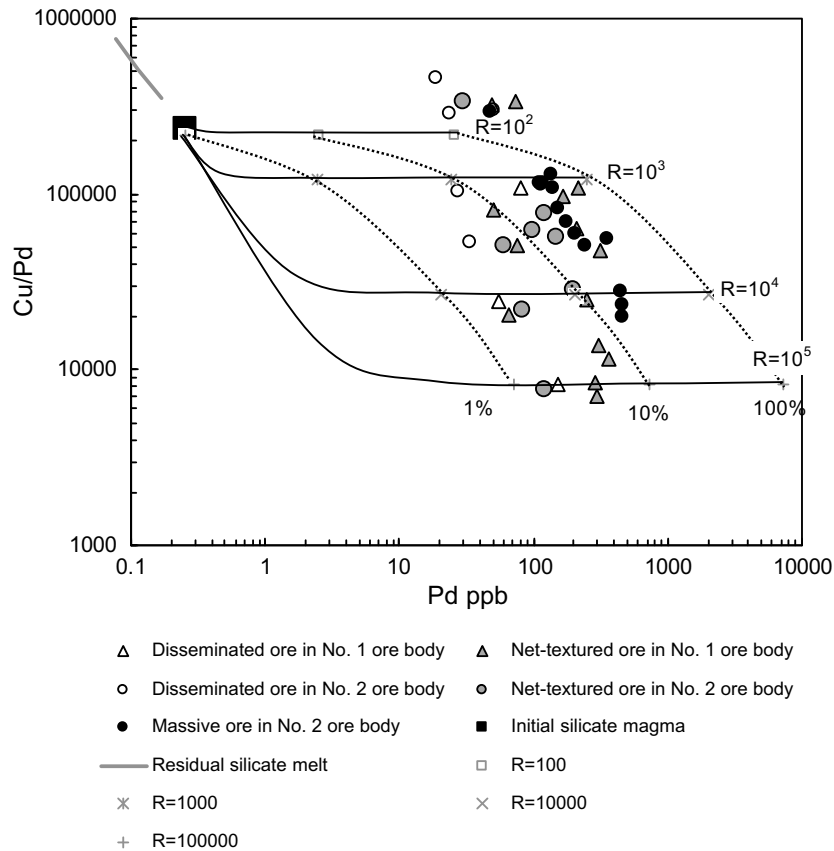


Fig. 10. Relationship between Cu/Pd values and Pd concentrations of the magmas and R factors for the Jinchuan sulfides. The magma initially contains 0.25 ppb Pd and 52 ppm Cu.

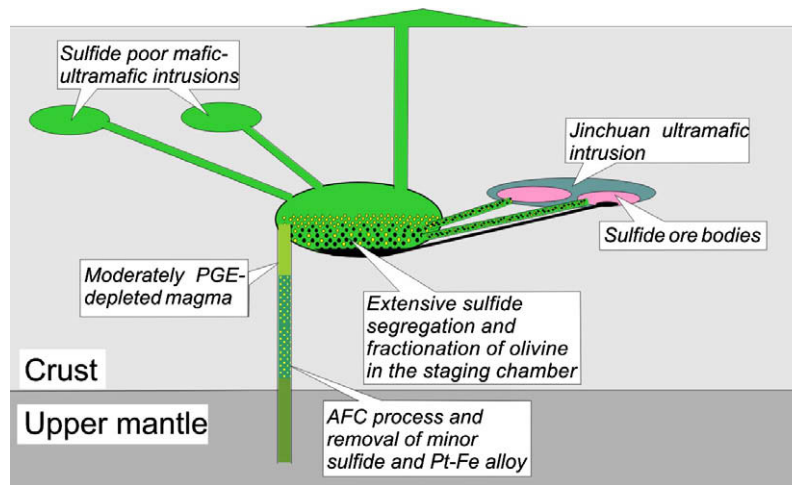


Fig. 11. Simplified model for the formation of the Jinchuan deposit.

magmas that were moderately depleted in the PGE but not Ni because of an earlier sulfide saturation event in the magma conduit system. A first stage sulfide saturation event may be also an important pre-requisite for the formation of other major Ni deposits such as Voisey's Bay and Pechenga, both of which have low PGE tenors (Ihlenfeld and

Keays, 2006). However, sulfides in the Noril'sk-Talnakh Ni-Cu-PGE deposits had segregated from magmas that had not undergone a prior S-saturated event (Lightfoot and Keays, 2005).

The strong PGE-depletion of the sulfide-free ultramafic rocks (Fig. 8A and Table 1) coupled with their strong crus-

tal contamination signature (Song et al., 2006) provides a strong exploration tool for Jinchuan-type Ni-Cu-(PGE) deposits. In contrast, the strongly PGE-depleted residual magmas are now represented the lower and middle Nadezhdinsky formation of the Siberia flood basalt sequence in the Noril'sk region (Brugmann et al., 1993; Lightfoot and Keays, 2005) and some of the other intrusions in the Nain Plutonic Suite, of which Voisey's Bay is one member (Keays and Lightfoot, 2008). Although the Sudbury, Canada "contact" sulfide ores are overlain by the PGE-depleted noritic rocks (as well as Ni and Cu), these Ni-Cu-PGE sulfides have been settled out of the overlying noritic rocks as the Sudbury impact melt sheet cooled (Lightfoot and Keays, 2005). Our model also proposes an exploration potential in the deep levels in the Jinchuan-style ore system.

6. CONCLUSIONS

1. The Jinchuan ultramafic intrusion and sulfide ore bodies in the middle portion of the intrusion were formed by injections of sulfide-free and sulfide-bearing olivine crystal mushes, respectively, from a deep-seated staging magma chamber.
2. The staging chamber was an open system. Extensive sulfide segregation occurred in the staging chamber and formed stratigraphic zones because of gravity settling of the sulfide droplets and olivine grains. When new pulses of magma intruded the magma chamber, the sulfide- and olivine-free magmas at the upper level were pushed out and formed sulfide-poor intrusions or lavas elsewhere.
3. Prior to entering the staging chamber, the magmas had experienced weak sulfide separation and hence loss of the PGE.
4. Massive sulfides at Jinchuan are strongly depleted in Pt. This may be because the magma batches that formed them had undergone fractionation of Pt-Fe alloys before intrusion into the staging chamber. It may also be because Pt was selectively leached from the sulfides by hydrothermal fluids during remobilization within the Jinchuan chamber.

ACKNOWLEDGMENTS

This study was supported by research grants from China (NSFC 40573014, 40730420, and "CAS Hundred Talents" Project from Chinese Academy of Sciences to X.-Y. Song) and a CRCG grant from the University of Hong Kong to M-F Zhou. Many thanks are due to Dr. Y.-L. Tian, Mr. Y.-S. Wang, D.-H. Ba, F.-G. Qiao, Y.-C. Ren, J.-P. Jiang, and S.-L. Zheng for their generous logistical support during the fieldwork. We are grateful to Professors C.M. Lesher and Nick Arndt for their thorough reviews that helped improve the paper and Professor E.W. Ripley for efficient editorial handling.

APPENDIX

The method for calculation of composition of metals in 100% sulfide liquid is

- (1) Assume that Ni in the sulfide-barren rocks is in olivine and deduce an equation between MgO and Ni in using the data of the lherzolite and dunite. Calculate how much Ni would enter olivine in the sulfide-lherzolite and sulfide-dunite by mean of MgO contents using this equation, and the excess Ni would be trapped by sulfide melt.
- (2) All of the Cu, and Zn combined with S to form chalcopyrite (CuFeS_2), and sphalerite (ZnS), and Ni in the sulfide melt form pentlandite ($\text{Fe}_{4.5}\text{Ni}_{4.5}\text{S}_8$).
- (3) Excess S combined with Fe to form pyrrhotite (Fe_9S_8) and pyrite (FeS_2), and any remaining Fe crystallized as Fe_2O_3 .

REFERENCES

- Arndt N. T., Czamanske G., Walker R. J., Chauvel C. and Fedorenko V. (2003) Geochemistry and origin of the intrusive hosts of the Noril'sk-Talnakh Cu-Ni-PGE sulfide deposits. *Econ. Geol.* **98**, 495–515.
- Arndt N. T., Lesher C. M. and Czamanske G. K. (2005) Mantle-derived magmas and magmatic Ni-Cu-(PGE) deposits. *Econ. Geol.* **100**, 5–24.
- Ballhaus C., Tredoux M. and Späth A. (2001) Phase relations in the Fe-Ni-Cu-PGE-S system at magmatic temperature and application to massive sulfide ores of the Sudbury Igneous Complex. *J. Petrol.* **42**, 1911–1926.
- Barnes S. J. and Tang Z. L. (1999) Chrome spinels from the Jinchuan Ni-Cu sulfide deposit, Gansu province, People's Republic of China. *Econ. Geol.* **94**, 343–356.
- Barnes S.-J. and Maier W. D. (1999) The fractionation of Ni, Cu, and the noble metals in silicate and sulfide liquids. In *Dynamic Processes in Magmatic Ore Deposits and their Application to Mineral Exploration* (eds. R. R. Keays, C. M. Lesher, P. C. Lightfoot and C. E. G. Farrow), Geological Association of Canada, Short course notes, 13, pp. 69–106.
- Bezmen N. I., Asif M., Brugmann G. E., Romanenko I. M. and Naldrett A. J. (1994) Distribution of Pd Rh Ru Ir Os and Au between sulfide and silicate melts. *Geochim. Cosmochim. Acta* **58**, 1251–1260.
- Bockrath C., Holzheid A. and Ballhaus C. (2004) Fractionation of the platinum-group elements during mantle melting. *Science* **305**, 1951–1953.
- Brugmann G. E., Naldrett A. J., Asif M., Lightfoot P. C., Gorbachev N. S. and Fedorenko V. A. (1993) Siderophile and chalcophile metals as tracers of the evolution of the Siberian Trap in the Noril'sk region, Russia. *Geochim. Cosmochim. Acta* **57**, 2001–2018.
- Cabri L. J. (1992) The distribution of trace precious metals in minerals and mineral products. *Mineral. Mag.* **56**, 289–308.
- Campbell I. H. and Naldrett A. J. (1979) The influence of silicate: sulfide ratios on the geochemistry of magmatic sulfides. *Econ. Geol.* **74**, 1503–1505.
- Campbell I. H. and Barnes S.-J. (1984) A model for the geochemistry of the platinum-group elements in magmatic sulfide deposits. *Can. Mineral.* **22**, 151–160.
- Chai G. and Naldrett A. J. (1992a) Characteristics of Ni-Cu-PGE mineralisation and genesis of the Jinchuan deposit, Northwest China. *Econ. Geol.* **87**, 1475–1495.
- Chai G. and Naldrett A. J. (1992b) The Jinchuan ultramafic intrusion: cumulate of a high-Mg basaltic magma. *J. Petrol.* **33**, 277–303.
- Czamanske G. K., Zen'ko T. E., Fedorenko V. A., Calk L. C., Budahn J. R., Bullock, Jr., J. H., Fries T. L., King B.-S.W. and

- Siems (1995) Petrographic and geochemical characteristics of ore-bearing intrusions of the Noril'sk type Siberia; with discussion of their origin. *Resource Geol. Special Issue* **18**, 1–48.
- De Waal S. A., Xu Z., Li C. and Mouri H. (2004) Emplacement of viscous mushes in the Jinchuan ultramafic intrusion, western China. *Can. Mineral.* **42**, 371–392.
- Ebel D. S. and Naldrett A. J. (1996) Experimental fractional crystallization of Cu- and Ni-bearing Fe sulfide liquids. *Econ. Geol.* **91**, 607–621.
- Fleet M. E. and Stone W. E. (1991) Partitioning of platinum-group elements in the Fe-Ni-S system and their fractionation in nature. *Geochim. Cosmochim. Acta* **53**, 245–253.
- Fleet M. E., Chrysosoulis S. I., Stone W. E. and Weisener C. G. (1993) Partitioning of platinum-group elements in the Fe-Ni-Cu-S system: experiments on the fractional crystallization of sulfide melt. *Contrib. Mineral. Petrol.* **115**, 36–44.
- Francis R. D. (1990) Sulfide globules in mid-ocean ridge basalts (MORB) and the effect of oxygen abundance in Fe-S-O on the ability of those liquids to partition metals from MORB and komatiitic magmas. *Chem. Geol.* **85**, 199–213.
- Hamlyn P. R. and Keays R. R. (1986) Sulphur saturation and second-stage melts: application to the Bushveld platinum metal deposits. *Econ. Geol.* **81**, 1431–1445.
- Ihlenfeld, C. and Keays, R. R. (2006) Sulfide segregation history of major magmatic Ni-Cu-(PGE) deposits: recognition of a two-stage process. *Aust Earth Science Conven*, July 2–7th, Melbourne.
- Keays R. R. and Crockett J. H. (1979) A study of precious metals in the Sudbury Nickel Irruptive ores. *Econ. Geol.* **65**, 438–450.
- Keays R. R., Nickel E. H., Groves D. I. and McGoldrick P. J. (1982) Iridium and palladium as discriminants of volcanic exhalative, hydrothermal and magmatic nickel sulfide mineralisation. *Econ. Geol.* **77**, 1535–1547.
- Keays R. R. (1995) The role of komatiitic and picritic magmatism and S-saturation in the formation of the ore deposits. *Lithos* **34**, 1–18.
- Keays R. R., Ihlenfeld C., McInnes B. I. A. and Zhou M.-F. (2004) Re-Os isotope dating of the Jinchuan Ni-Cu-PGE sulfide deposit, China. In *Recent Advances in Magmatic Ore Systems of Mafic-Ultramafic Rocks, Proceedings of the IGCP 479*, (eds. J. G. Shellnutt, M.-F. Zhou and K. N. Peng), pp. 41–42.
- Keays R. R. and Lightfoot P. C. (2007) Siderophile and chalcophile metal variations in Tertiary picrites and basalts from West Greenland with implications for the sulfide saturation history of continental flood basalt magmas. *Mineral. Deposita* **42**, 319–336.
- Keays R. R. and Lightfoot P. C. (2008) *Platinum Group Elements in the Nain Plutonic Suite: Implications for the Genesis of the Voisey's Bay Ni-Cu-Co Sulphide Deposit*. GAC-MAC conference, May 24–28, Quebec City.
- Kullerud G., Yund R. A. and Moh G. (1969) Phase relations in the Fe-Ni-S Cu-Fe-S and Cu-Ni-S systems. *Econ. Geol. Monogr.* **4**, 323–343.
- Latypov R. (2007) Noril'sk- and Lower Talnakh-type intrusions are not conduits for overlying flood basalts: insights from residual gabbroic sequence of intrusions. *Appl. Earth Sci.* **116**, 215–225.
- Lehmann J., Arndt N., Windley B., Zhou M.-F., Wang C. Y. and Harris C. (2007) Field relationships and geochemical constraints on the emplacement of the Jinchuan Intrusion and its Ni-Cu-PGE Sulfide Deposit, Gansu, China. *Econ. Geol.* **102**, 75–94.
- Leshner C. M. and Keays R. R. (1984) Metamorphically and hydrothermally mobilised Fe-Ni-Cu sulfides at Kambalda, Western Australia. In *Sulfide deposits in mafic and ultramafic rocks* (eds. D. L. Buchanan and M. J. Jones), pp. 62–69, Inst. Min. Metall. (London).
- Leshner C. M. and Stone W. E. (1996) Exploration geochemistry of komatiites. In *Trace Element Geochemistry of Volcanic Rocks: Applications for Massive Sulfide Exploration* (ed. D. A. Wyman), Geological Association of Canada short Course Notes, 12, 153–204.
- Leshner C. M. and Keays R. R. (2002) Komatiite-associated Ni-Cu-(PGE) deposits. In *The Geology, Geochemistry, Mineralogy, Mineral Beneficiation of the Platinum-Group Elements* (ed. L. J. Cabri), Canadian Institute of Mining, Metallurgy and Petroleum, 54, 579–617.
- Li C., Naldrett A. J., Coats C. J. A. and Johannessen P. (1992) Platinum, palladium, gold and copper-rich stringers at Strathcona mine, Sudbury: their enrichment by fractionation of a sulfide liquid. *Econ. Geol.* **87**, 1584–1596.
- Li C. and Naldrett A. J. (1999) Geology and petrology of the Voisey's Bay intrusion: relation of olivine with sulfide and silicate liquids. *Lithos* **47**, 1–31.
- Li C., Xu Z., Sybrand A., De Waal S. A., Ripley E. M. and Maier W. D. (2004) Compositional variations of olivine from the Jinchuan Ni-Cu sulfide deposit, western China: implication for ore genesis. *Mineral. Deposita* **39**, 159–172.
- Li X. H., Su L., Chung S.-L., Li Z. X. and Liu Y. (2005) Formation of the Jinchuan ultramafic intrusion and the world's third largest Ni-Cu sulfide deposit: associated with the 825 Ma south China mantle plume? *Geochem. Geophys. Geosyst.* **6**, 1–16.
- Lightfoot P. C. and Naldrett A. J. (1999) Geological and geochemical relationships in the Voisey's Bay Intrusion, Nain Plutonic Suite, Labrador, Canada. In *Dynamic Processes in Magmatic Ore Deposits and their Application in Mineral Exploration* (eds. R. R. Keays, C. M. Leshner, P. C. Lightfoot and C. E. G. Farrow), Ottawa, Geological Association of Canada, Short Course vol. 13, 1–30.
- Lightfoot P. C. and Keays R. R. (2005) Siderophile and chalcophile metal variations in flood basalts from the Siberian Trap, Noril'sk Region: implications for the origin of the Ni-Cu-PGE sulfide ores. *Econ. Geol.* **100**, 439–462.
- Momme P., Tegner C., Brooks C. K. and Keays R. R. (2002) The behaviour of platinum-group elements in basalts from the East Greenland rifted margin. *Contrib. Mineral. Petrol.* **143**, 133–153.
- Naldrett A. J., Innes D. G., Sowa J. and Gorton M. (1982) Compositional variation within and between 5 Sudbury ore deposits. *Econ. Geol.* **77**, 1519–1534.
- Naldrett A. J., Lightfoot P. C., Fedorenko V. A., Gorbachev N. S. and Doherty W. (1992) Geology and geochemistry of intrusions and flood basalts of the Noril'sk region, USSR, with implications for the origin of the Ni-Cu-PGE ores. *Econ. Geol.* **87**, 975–1004.
- Naldrett A. J., Fedorenko V. A., Lightfoot P. C., Kunilov V. E., Gorbachev N. S., Doherty W. and Johan J. (1995) Ni-Cu-PGE deposits of the Noril'sk region Siberia: their formation in conduits for flood basalt volcanism. *Trans. Inst. Min. Met.* **104**, B18–B36.
- Naldrett A. J. and Lightfoot P. C. (1999) Ni-Cu-PGE deposits of the Noril'sk region, Siberia; their formation in conduits for flood basalt volcanism. In *Dynamic Processes in Magmatic ore Deposits and their Application in Mineral Exploration* (eds. R. R. Keays, C. M. Leshner, P. C. Lightfoot and C. E. G. Farrow), Ottawa, Geological Association of Canada, Short Course vol. 13, 195–249.
- Naldrett A. J., Asif M., Krstic S. and Li C. (2000) The composition of ore at the Voisey's Bay Ni-Cu sulfide deposit with special reference to platinum-group elements. *Econ. Geol.* **95**, 845–866.

- Naldrett A. J. (2005) *Magmatic Sulfide Deposits – Geology, Geochemistry and Exploration*. Springer, Berlin, Heidelberg, New York.
- Nixon G. T. and Hammack J. L. (1991) Metallurgy of ultramafic–mafic rocks in British Columbia with emphasis on the platinum-group elements. In *Ore Deposits, Tectonics and Metallogeny in the Canadian Cordillera. Energy, Mines and Petroleum Resources*, 125–162, British Columbia.
- Peach C. L., Mathez E. A. and Keays R. R. (1990) Sulfide melt-silicate melt distribution coefficients for noble metals and other chalcophile elements as deduced from MORB: implication for partial melting. *Geochim. Cosmochim. Acta* **54**, 3379–3389.
- Peach C. L., Mathez E. A., Keays R. R. and Reeves S. J. (1994) Experimentally determined sulfide melt-silicate melt partition coefficients for Ir and Pd. *Chem. Geol.* **117**, 361–377.
- Peck D. and Keays R. R. (1990a) Insights into the behaviour of the precious metals in primitive, S-undersaturated magmas: evidence from the Heazlewood river complex, Tasmania. *Can. Mineral.* **28**, 553–578.
- Peck D. C. and Keays R. R. (1990b) Geology, geochemistry and origin of platinum group element – chromitite occurrences in the Heazlewood River Complex, Tasmania. *Econ. Geol.* **85**, 765–793.
- Qi L., Zhou M.-F. and Wang C. Y. (2004) Determination of low concentrations of platinum group elements in geological samples by ID-ICP-MS. *J. Anal. Atomic Spectrom.* **19**, 1335–1339.
- Qi L. and Zhou M.-F. (2008) Platinum-group elemental and Sr–Nd–Os isotopic geochemistry of Permian Emeishan flood basalts in Guizhou Province, SW China. *Chem. Geol.* **248**, 83–103.
- Rajamani V. and Naldrett A. J. (1978) Partitioning of Fe, Co, Ni, and Cu between sulfide liquid and basaltic melt and the composition of Ni–Cu sulfide deposits. *Econ. Geol.* **73**, 82–93.
- Redman B. A. and Keays R. R. (1985) Archaean basic volcanism in the Eastern Goldfields Province, Yilgarn Block, Western Australia. *Precam. Res.* **30**, 113–152.
- Ripley E. M., Lightfoot P. C., Li C. and Elswick E. R. (2003) Sulfur isotopic studies of continental flood basalts in the Noril'sk region: implications for the association between lavas and ore-bearing intrusions. *Geochim. Cosmochim. Acta* **67**, 2805–2817.
- Ripley E. W., Sarkar A. and Li C. (2005) Mineralogic and stable isotope studies of hydrothermal alteration at the Jinchuan Ni–Cu deposit, China. *Econ. Geol.* **100**, 1349–1361.
- Rowell W. F. and Edgar A. D. (1985) Genesis of platinum-group elements and gold in a hydrothermal sulfide occurrence in a Nipissing intrusion, northern Ontario. *Can. Mineral.* **23**, 313.
- Sixth Geological Unit. (1975) Exploration report of the Baijiazui Cu–Ni deposit, Sixth Geological Unit, Geological Survey of Gansu Province.
- Sixth Geological Unit. (1984) Geology of the Baijiazui Cu–Ni sulfide deposit, Sixth Geological Unit, Geological Survey of Gansu Province, Geological publishing House, Beijing.
- Song X.-Y., Zhou M.-F., Wang C. Y. and Qi L. (2006) Role of crustal contamination in formation of the Jinchuan intrusion and its world-class Ni–Cu–(PGE) sulfide deposit, NW China. *Int. Geol. Rev.* **48**, 1113–1132.
- Song X.-Y., Zhou M.-F., Tao Y. and Xiao J.-F. (2008) Controls on the metal compositions of magmatic sulfide deposits in the Emeishan large igneous province, SW China. *Chem. Geol.* **253**, 38–49.
- Stone W. E., Crocket J. H. and Fleet M. E. (1990) Partitioning of palladium, iridium, platinum and gold between sulfide liquid and basalt melt at 1200 °C. *Geochim. Cosmochim. Acta* **54**, 2341–2344.
- Stumpfl E. F. and Rucklidge J. (1982) The platiniferous dunite piped of the eastern Bushveld. *Econ. Geol.* **77**, 1419–1431.
- Tang Z. L. (1991) Formation of the Jinchuan Cu–Ni sulfide deposit. *Modern Geol.* **4**, 55–64, in Chinese.
- Tang Z. L. and Li W. Y. (1995) *Mineralisation Model and Geology of the Jinchuan Ni–Cu Sulfide Deposit Bearing PGE*. Geological Publishing House, Beijing.
- Vogel D. C. and Keays R. R. (1997) The petrogenesis and platinum-group element geochemistry of the Newer Volcanic Province, Victoria, Australia. *Chem. Geol.* **136**, 181–204.
- Yang J.-S., Liu F.-L., Wu C.-L., Xu Z.-Q., Shi R.-D. and Chen S.-Y. (2005) Two ultrahigh-pressure metamorphic events recognized in the central orogenic belt of China: evidence from the U–Pb dating of coesite-bearing zircons. *Int. Geol. Rev.* **47**, 327–343.
- Zhou M.-F., Yang Z.-X., Song X.-Y., Keays R. R. and Lesher C. M. (2002) Magmatic Ni–Cu–(PGE) sulfide deposits in China. In *The Geology, Geochemistry, Mineralogy, Mineral Beneficiation of the Platinum-Group Elements*. Canadian Institute of Mining, Metallurgy and Petroleum (ed. L. J. Cabri) Special vol. 54, 619–636.

Associate editor: Edward M. Ripley